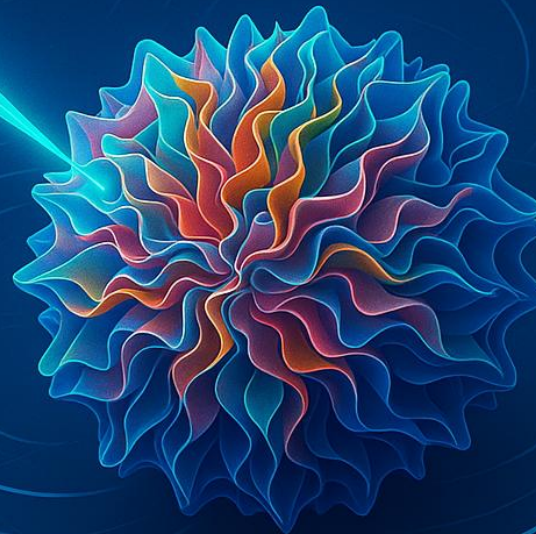


RADEXEL
EXCELLENCE IN ONCOLOGY

The Most Precise Radiation Therapy Solution

2025.12



Confidential - Do Not Distribute

Highlights

Problem

Despite \$100B spent annually on Radiation Therapy (RT), **patients still suffer from adverse effects.**

Solution

MPRT reduces skin & mucosal toxicities by up to 50%, as shown in pilot clinical studies.

Market

Retrofittable to 17,000 RT systems worldwide, enabling rapid adoption and global expansion.

Target Problem

Solution: MPRT-SD

Team & Execution

Market & Business

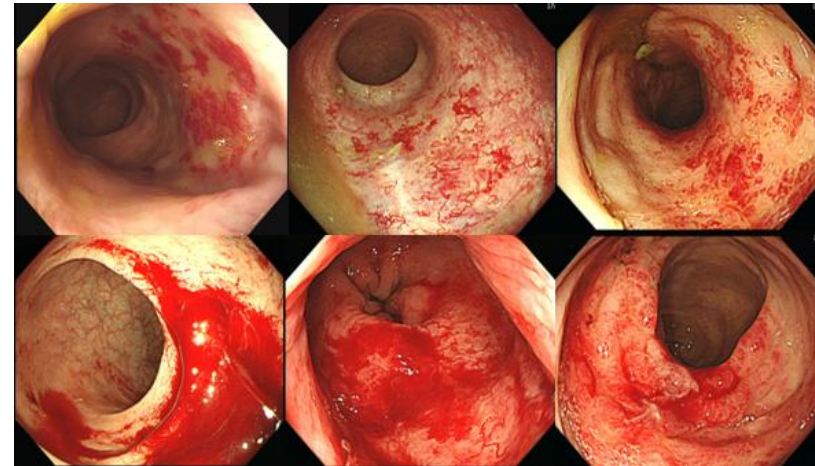
Expansion: MPRT-IS & MPRT-RB

Target Problem

A Key Limitation of Radiation Therapy (RT): Damage to Normal Tissue Surrounding the Tumor



Damage to Skin: breast cancer, H&N cancer, etc.

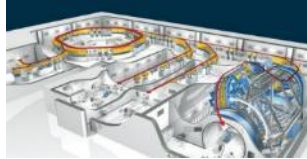


Damage to Mucosa: prostate cancer, oral cavity, etc.

Demand for Improved RT

X-ray Therapy: Core Patents Expired, Competition Intensifying

Particle Therapy: High Cost with Limited Reduction in Side Effects

	 X-ray therapy	 Proton therapy	 Carbon ion therapy
Tx efficacy	Standard Tx	Improved Tx in limited indications	
Manufacturers	Siemens/Varian (US) Elekta (Sweden) Accuray (US) Shinva (China) United imaging (China)	IBA (Belgium) Siemens/Varian (US) Hitachi (Japan) Mevion (US)	Hitachi (Japan) Toshiba (Japan)
Equipment price	\$ 5 - 7 M	\$ 30 – 70 M	\$ 400 M
Installed base	17,000	100	14

Target Problem

Solution: MPRT-SD

Team & Execution

Market & Business

Expansion: MPRT-IS & MPRT-RB

Technical Background

Mechanism of X-ray Therapy: DNA Damage Caused by Secondary Electrons

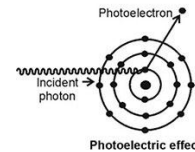
X-ray Irradiation



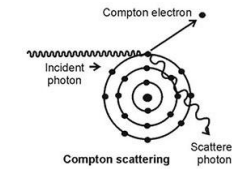
X-rays are delivered to the body.

Secondary Electron Generation

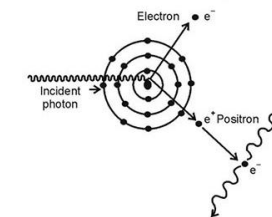
Photoelectric effect



Compton scattering

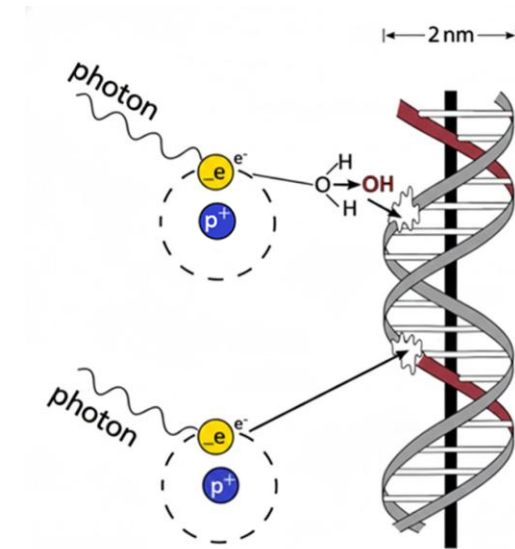


Pair production



The X-rays interact with atoms in the device or the body to emit secondary electrons.

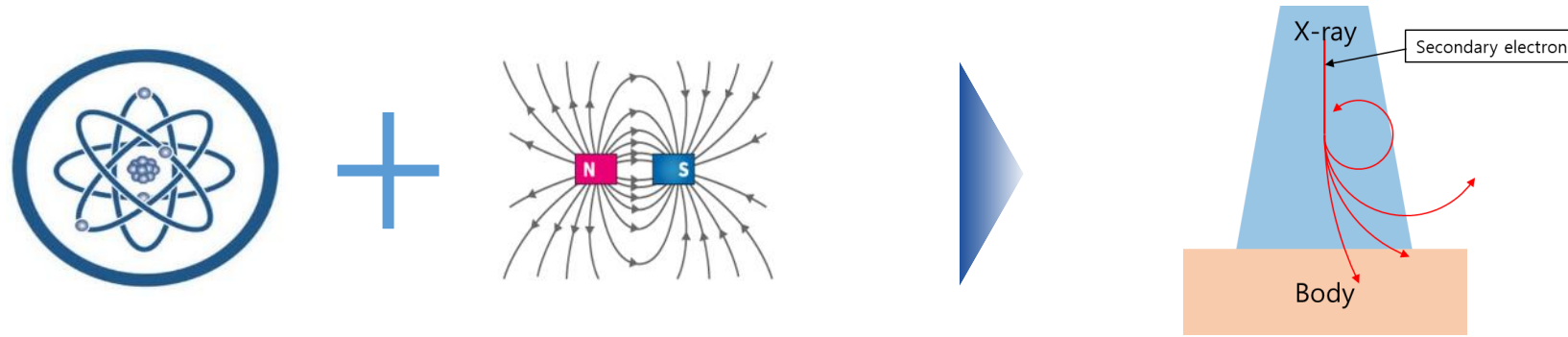
DNA Damage by Electrons



The electrons damage cells by breaking DNA directly or indirectly.

Technical Concept

Magnetic Control Can Adjust Electron Paths and Radiation Dose Distribution



Magnetic Precision Radiation Therapy, MPRT

Product Pipeline

Magnetic Precision Radiation Therapy, MPRT



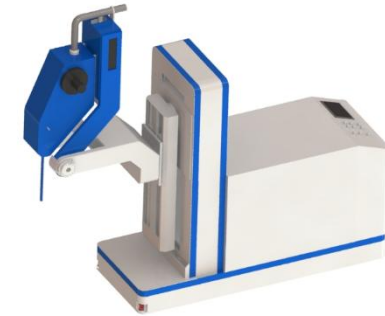
MPRT-SD

Modular MPRT for Standard RT
Fast Adoption & Near-term Revenue



MPRT-IS

Integrated MPRT System
Long-term Differentiation



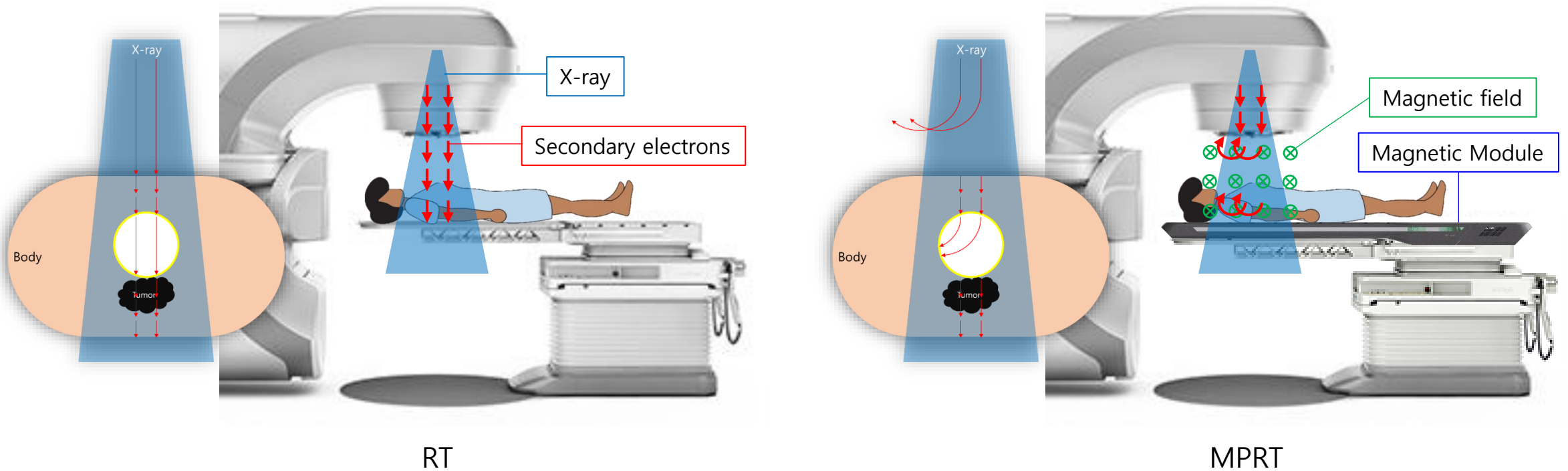
MPRT-RB

Mobile MPRT Robotic System
Next-Generation Platform

MPRT-SD, Mechanism

Tissue–Air Density Contrast + Magnetic Field Control

→ Minimizes Skin and Mucosal Toxicities



MPRT-SD, R&D

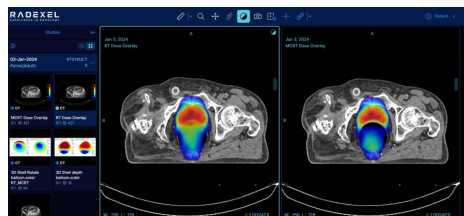
MPRT-SD: Integrating Couch Module, Balloon Applicator, and Planning Software



MPRT-CM (Couch Module)



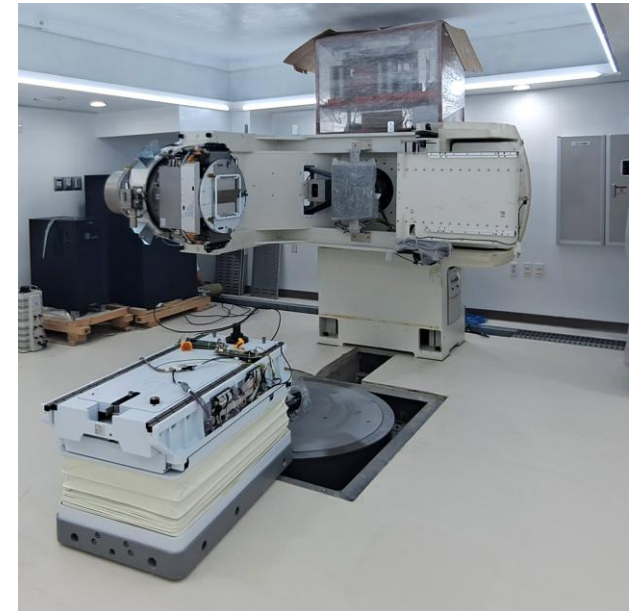
MPRT-BA (Balloon Applicator)



MPRT-PS (Planning Software)



MPRT System



X-ray RT device

MPRT-SD, R&D

MPRT-SD: Currently in pilot clinical use



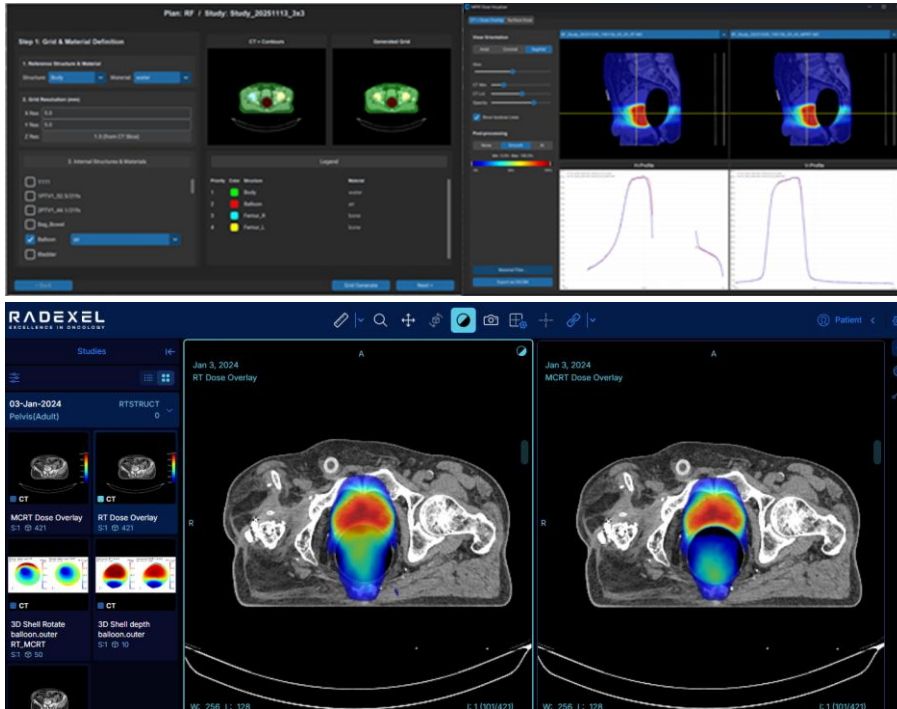
MPRT-CM (Couch Module)



MPRT-CS (Control Software)

MPRT-SD, R&D

MPRT-PS (Planning Software): Treatment planning with Magnetic field integration



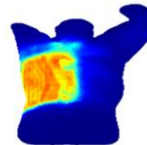
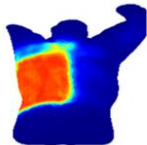
For clinicians:
Cloud-based MPRT planning platform

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홈 > 목록 > 우측 유방암 방사선치료(MPRT) 안내

우측 유방암 방사선치료(MPRT) 안내 진료 요약

08.23.2025
Kangwon Organization | Nuri Jung



일반 방사선치료 **자기 정밀 방사선치료 (MPRT)**

MPRT 기술 적용

- 자기장 제어를 이용해 방사선의 경로와 분포를 정밀하게 조절하는 기술입니다.
- 일반 방사선치료에 비해 정확도가 높고, 정상조직의 손상을 줄일 수 있습니다.

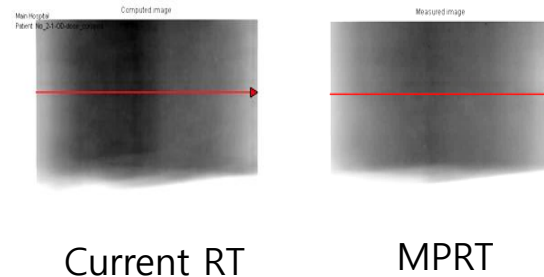
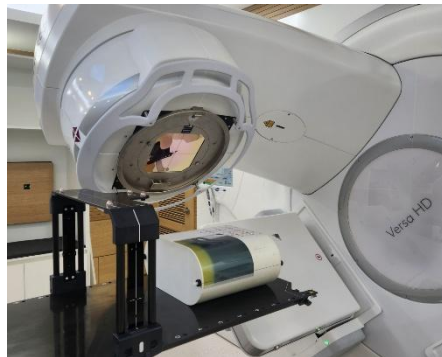
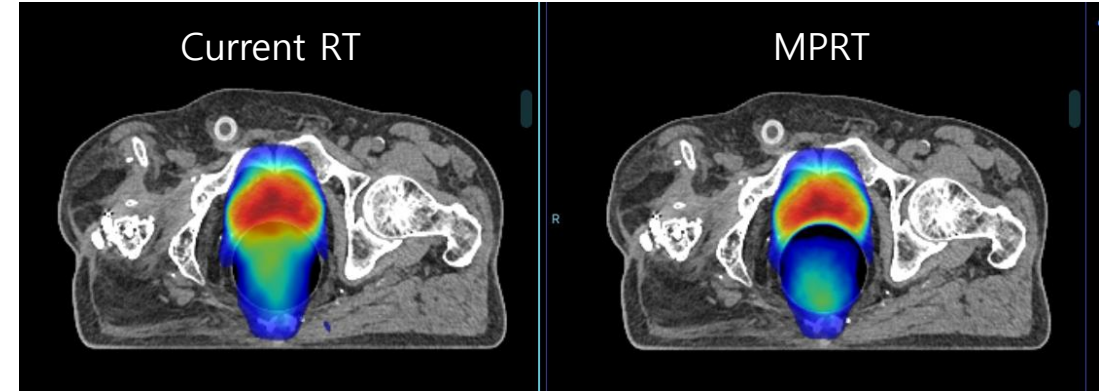
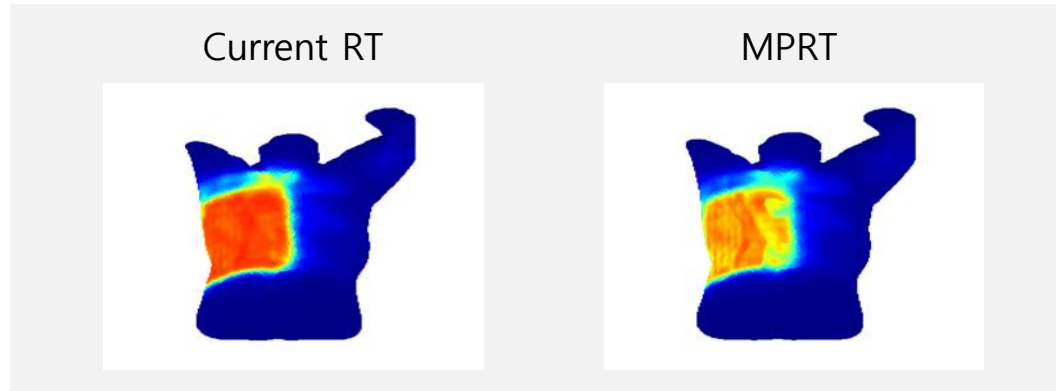
MPRT의 효과

- 피부에 노출되는 방사선량이 최대 25%, 평균 15% 감소됩니다.
- 방사선에 의한 피부 부작용 (색소 침착, 건조 등) 저감이 예상됩니다.
- 유방암 치료 방사선량은 3% 이내의 오차로 관리됩니다.

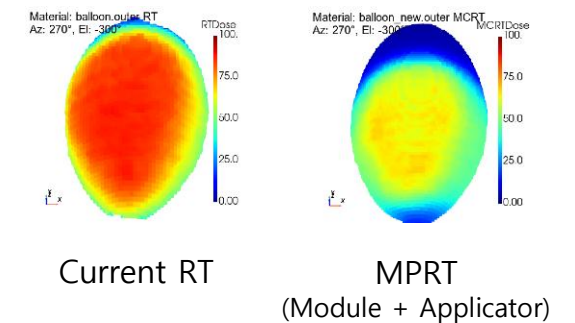
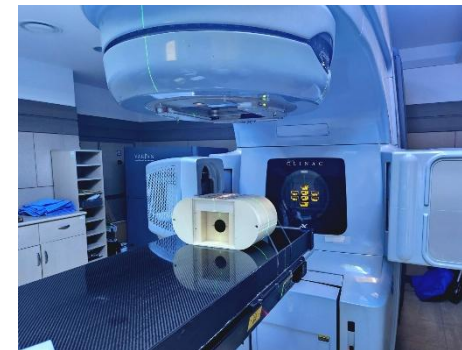
For patients:
Mobile app for treatment guidance

Preclinical Validation

Phantom Study: 15–30 % reduction in normal tissue dose



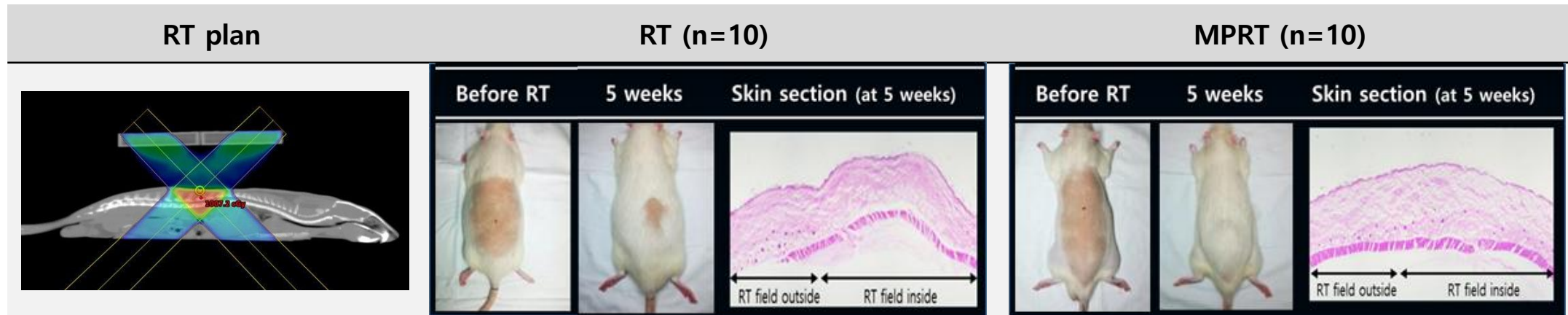
Skin dose: 10-20 % ↓



Mucosal dose: 25-35 % ↓

Preclinical Validation

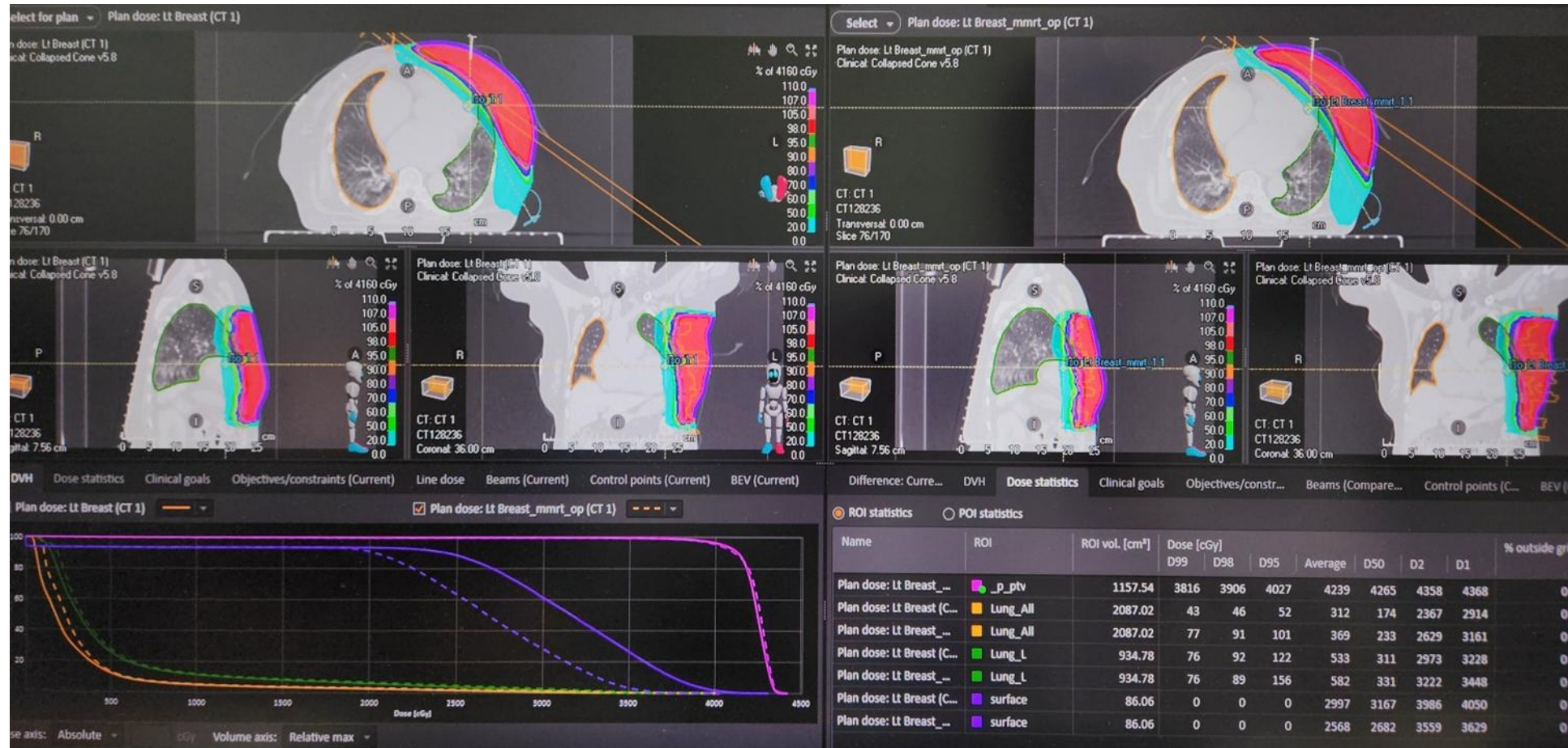
Animal Study: Hair loss reduced from 80% to 14% (skin-sparing effect)



- At 5 weeks, mean in-field hair loss rate was 80.9 % (range 48.2 - 100) without magnetic field, and 13.9 % (range 0.9 – 29.4) with magnetic field
- MPRT reduced radiation-induced skin toxicity.

Clinical Trial: Breast cancer

Treatment Plan: Maintained target dose with ~15% reduction in skin dose



Clinical Trial: Breast cancer

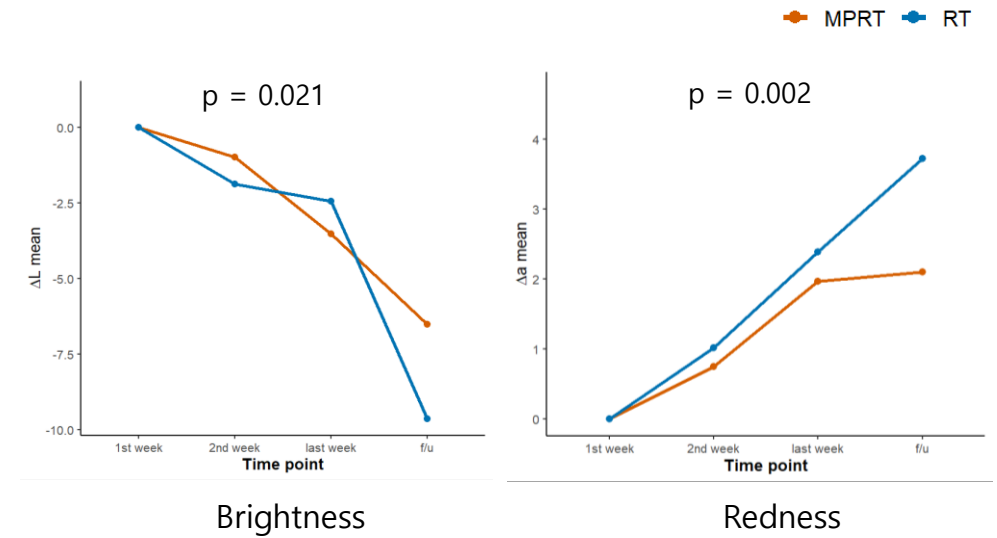
Patient Evaluation: Reduced skin pigmentation and erythema by ~50% ($p < 0.05$)

Radiation dermatitis after RT and MPRT



Severity of skin pigmentation

	RT (n=11)	MPRT (n=36)
Mild	18.2 %	61.1 %
Moderate	45.5 %	27.8 %
Severe	36.4 %	11.1 %



MPRT-SD, Target Indications

Minimizes Skin and Mucosal Damage



MPRT-SD



Dermatitis, breast



Dermatitis, neck



Mucositis, rectum



Alopecia

Target Problem

Solution: MPRT-SD

Team & Execution

Market & Business

Expansion: MPRT-IS & MPRT-RB

Team: Core Management

Experienced Founders with Proven Expertise in Oncology and Engineering



Kim Tyson, MD, CEO, Co-founder
Former CEO of AI medtech
Strategy and Scaling



Jung Nuri Hyun, MD, CMO, Co-founder
Radiation Oncologist
R&D and Validation



Bhang Jung-bae, PhD, CTO
Accelerator physicist
System Design

Team: Key Specialists & Advisors

Key specialists



Jung Su-hyun, MBA, CFO
Business administration
Finance and HR



Nam Seung-hee, PhD
Computer engineering
Accelerator and Robotics control



Kim Tae-hoon, PhD
Nuclear engineering
Particle simulation, Medical physics



Song Ye-eun
Medical engineering
Regulatory affairs, 13 years



Lee Sang-gil, PhD
Computer engineering
TPS development



Kim Sung-jun
Mechanical engineering
Medical device design, 13 years

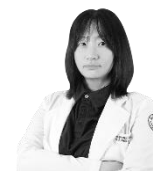
Advisors from top cancer centers



Song Si-yeol, MD,
Director of Cancer Hospital



Yoo Do-sol, MD
Director of Department



Choi Yoon-sun, MD
Professor



Lim Chae-hong, MD
Professor



Yoo Gyu-sang, MD
Professor

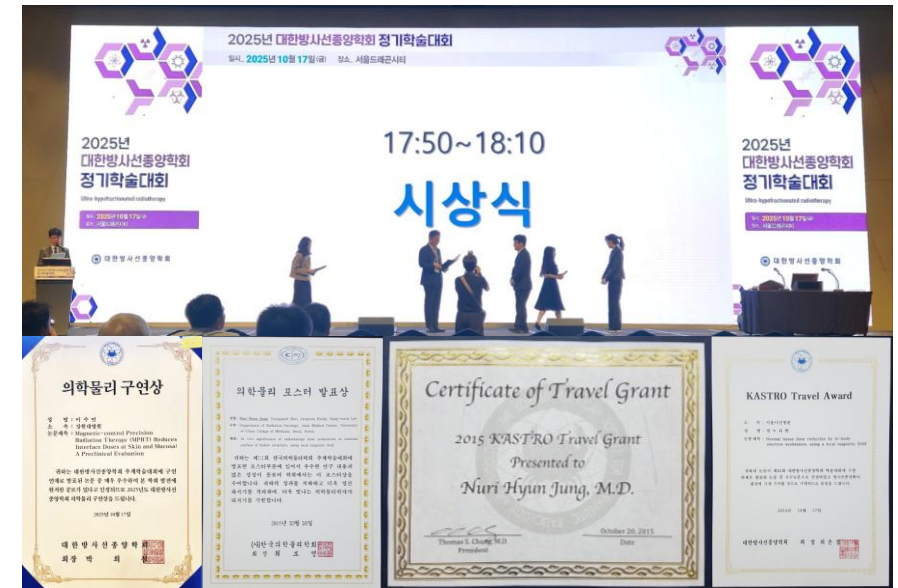


Yoo Je-sang, MD
Professor

Traction & Recognition

Validated and Trusted by the Scientific and Clinical Communities

- ✓ **\$6M** total capital raised
- ✓ **10 radiation oncology professors** among investors
- ✓ **National R&D grants** \$3M secured
- ✓ **Awards** from KOSRO, KSMP, KASTRO
- ✓ Ongoing clinical trials (breast & prostate cancer)

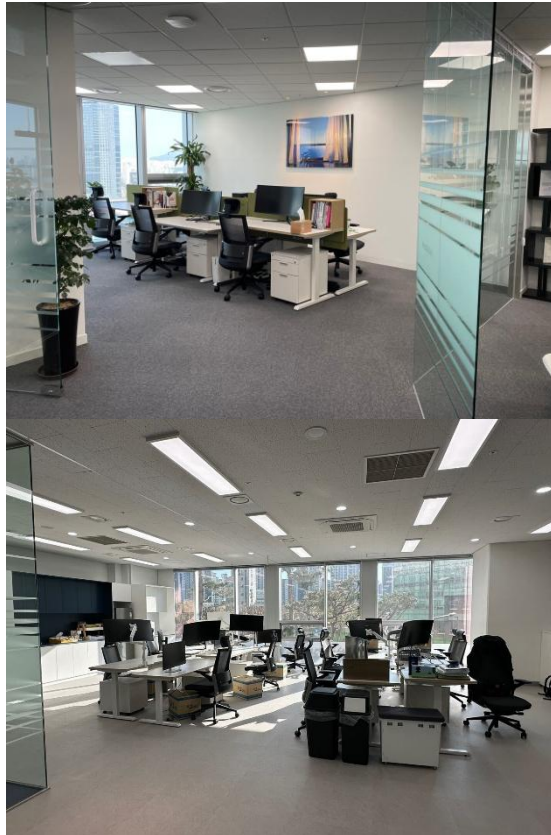


R&D Facilities

Equipped for R&D, Manufacturing, and Clinical Trials



HW Development



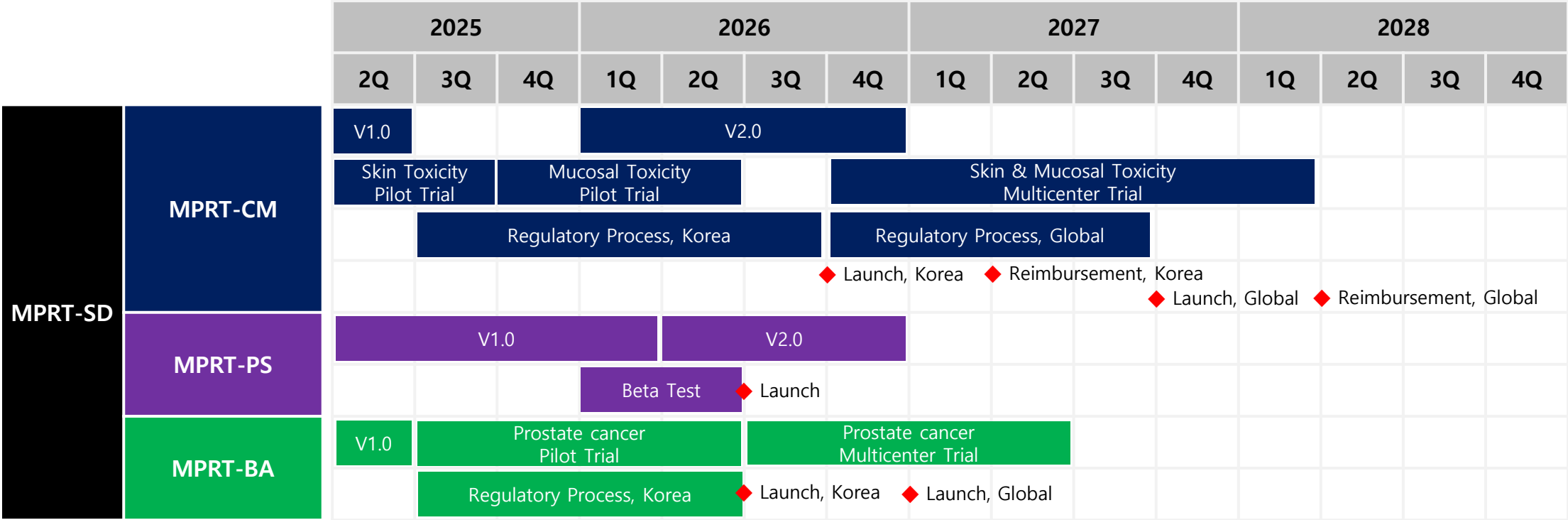
SW Development, RA



Clinical Research

MPRT-SD Product Roadmap

Product development completed; clinical validation in progress



Securing IP

72 Patents Filed/Granted · FTO Analysis Completed

	Filed	Granted
Korea	5	8
US	3	6
Europe	5	1
China	5	1
Japan	7	2
India	6	0
Canada	4	0
Australia	4	2
Russia	1	3
Singapore	3	0
South Africa	0	1
Mexico	1	0
Vietnam	1	0
Indonesia	1	1
Philippines	1	0
Total	47	25



EXCELLENCE IN ONCOLOGY

자기관-방사선 융합 치료 시스템 FTO 분석
및 지재산 보완 방안 분석

작성자 : 정태균 파트너 변호사(Karl@BLT.kr)

The only firm for your success
BLT IP & Law Firm

info@BLT.kr BLT.kr 02.514.0104

결론(2/2)

4. 침해 우려 낮음 특허군

- #1, #3, #4, #5, #6, #7, #8, #9, #10
- 공자선 기반이 아니거나, 겹트라 입자선 MRI 연동 등을 포함하여 직접 대응되지 않음.
- 전용실시간 보완 특허(#2, #4, #5)는 자유 이용 가능.

5. 상세분석 필요 특허군

- #2, #12, #14, #20-#33
- 일부 기능(자기관 제어 전자기동 보정)이 유사함.
- 대상 기술의 핵심 목적: 보정을 소프트웨어 제어
- 선행기술의 목적: 선량 증가 또는 입자선 제어
- 선행특허는 상시과정에서 기능 한정(disparate, focus)
- 한정된 구성 범위와 대상 기술의 실제 구현이 불일치, 실제 침해 위험은 제한적.

6. 결론적 위험 수준

- 전체적으로 법적 침해 리스크는 낮음.

8. 결론적 위험 수준

- 전체적으로 법적 침해 리스크는 낮음.
- 단, 소프트웨어 기반 자체제어 로직이 주요 차별점임으로,
- GUI 제어, preset 관리 방식
- 실시간 자식-Couch-이소센터 좌표계 연동부분은 지속적인 모니터링 및 방어적 특허 확보 필요.

Target Problem

Solution: MPRT-SD

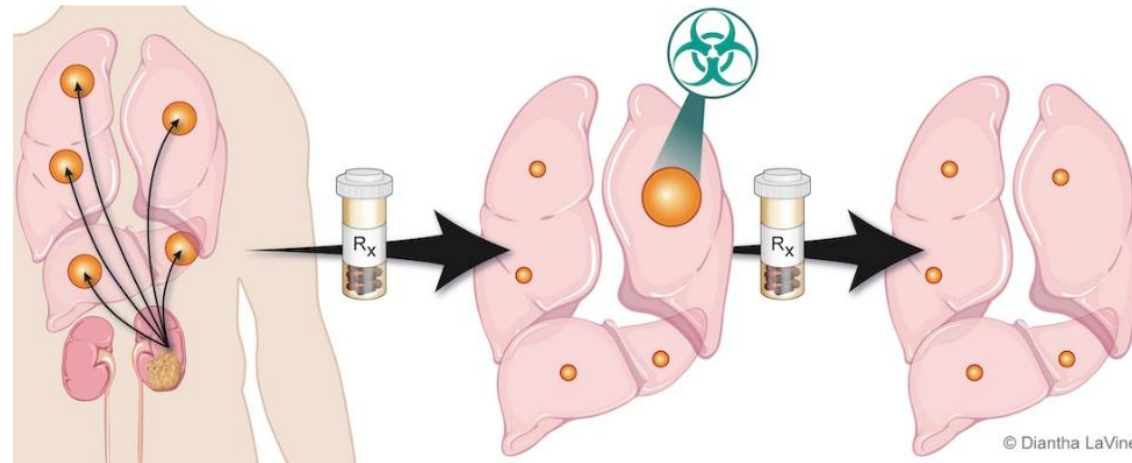
Team & Execution

Market & Business

Expansion: MPRT-IS & MPRT-RB

Market Driver

As cancer incidence and survival rates increase, demand for repeated RT is rising.



As survival improves, patients live long enough to require additional radiation therapy, especially when lesions recur or become resistant after systemic therapy.

Market Size

>\$100B Global Radiation Therapy Market

- **Market Fundamentals**

- **17,000** installed RT systems worldwide
- **~250** patients treated per device per year (US 250, Korea 400)
- **~\$25K** average treatment cost per patient (US \$50K, Korea \$10K)

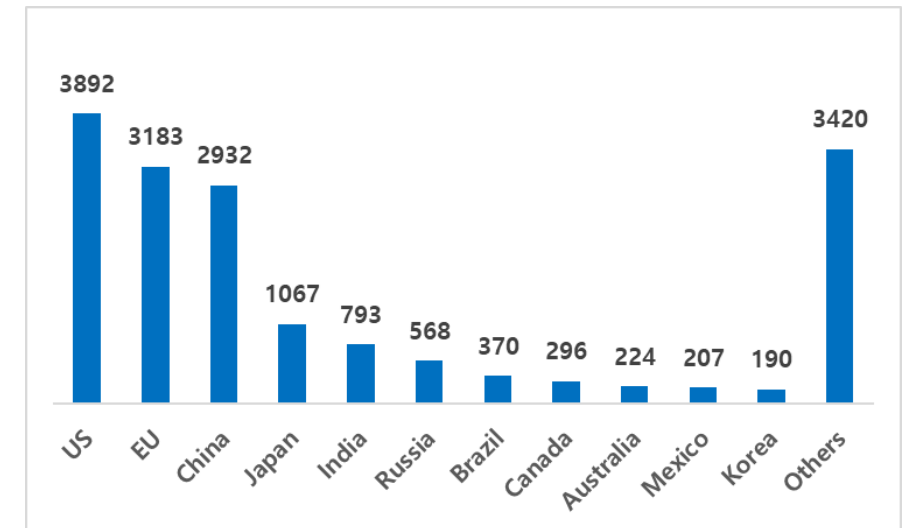
- **Estimated Global Market Size**

- **250 patients × 17,000 devices × \$25K = ~\$106B per year**

- **Key Insight**

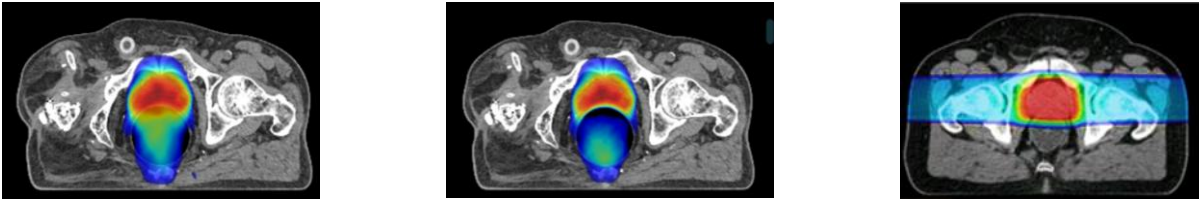
- Radiation therapy represents a **>\$100B annual global market**, driven by rising cancer incidence and increasing adoption of advanced RT technologies.

Number of RT Devices by Region



Competitive Advantage

Significant normal tissue dose reduction, without particle-level cost

			
	Standard RT	MPRT-SD	Particle Therapy (Proton/Carbon ion)
Type	Standalone	Add-on Module	Standalone
Treatment Revenue (per patient)	\$15-50K	+\$5K (incremental)	\$50-100K
Upfront Capital Cost (per device)	\$5-7M	\$0 (revenue share) \$1.5M (upfront payment)	\$30-400M
Advantages	-	Reduces side effects (breast, prostate, H&N)	Reduces side effects (brain, pediatric)
Disadvantages	Higher normal tissue injury	-	High upfront cost

- Comparable toxicity-reduction procedure (SpaceOAR): \$6–9K per patient

Business Model - Revenue Share

Aligned incentives for hospitals and Radexel

No. of RT Patients per Device	250
Assumed MPRT Adoption Rate (%)	50
No. of MPRT Patients per Device	125
MPRT Fee per Patient (\$K)	5
Total Revenue per Device (\$K)	625
Revenue Split (Hospital : Radexel)	50:50
Hospital Revenue Share (\$K)	313
Radexel Revenue Share (\$K)	313

- **Hospitals** adopt MPRT with **zero upfront cost**
- **Radexel** benefits from long-term **per-patient revenue sharing**
- **~74% of RT patients are addressable** by MPRT in the U.S.
(indication coverage: breast, prostate, H&N, uterus, brain)
- Revenue estimates **assume a conservative 50% adoption**,
reflecting country- and hospital-level variability

Business Model - Upfront Payment

Hospital IRR Reaches 31.4% with \$1.5M Installation and \$0.15M Annual Fee

Estimated Cash Flow for Hospitals Adopting MPRT-SD

	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10	SUM
Revenue (\$K)	-	375	750	750	750	750	750	750	750	750	750	7,125
No. of RT Patients	250	250	250	250	250	250	250	250	250	250	250	
MPRT Adoption (%)	0%	30%	60%	60%	60%	60%	60%	60%	60%	60%	60%	
No. of MPRT Patients	-	75	150	150	150	150	150	150	150	150	150	
Tx. Fee per Patient (\$K)	-	5	5	5	5	5	5	5	5	5	5	
Cost (\$K)	1,500	150	150	150	150	150	150	150	150	150	150	3,000
Equipment Purchase	1,500	-	-	-	-	-	-	-	-	-	-	
SW/Applicator/Mainten.	-	150	150	150	150	150	150	150	150	150	150	
Profit (\$K)	- 1,500	225	600	600	600	600	600	600	600	600	600	4,125

- **~74% of RT patients are addressable** by MPRT in the U.S. (indication coverage: breast, prostate, H&N, uterus, brain)

Market Adoption Trajectory

Conservative MPRT adoption modeled against IMRT precedent

Year	No. of RT devices	No. of MPRT	Cumulative No. of MPRT
2027	17,000	5	5 (0.0%)
2028	17,459	17	22 (0.1%)
2029	17,930	63	85 (0.5%)
2030	18,415	151	236 (1.3%)
2031	18,912	362	600 (3.2%)
2032	19,422	870	1,470 (7.6%)
2033	19,947	1,000	2,470(12.4%)
2034	20,485	1,000	3,470 (16.9%)
2035	21,038	1,000	4,470 (21.2%)

- Assumes ~2.7% annual growth in installed RT devices
- **Adoption assumptions are intentionally conservative** and do not reflect potential acceleration from zero-capex deployment or OEM partnerships.

Comparable adoption trajectory

- **IMRT Example**

- Launch module in 1999; Reimbursement in 2000;
- **32% adoption by 2003**; 87% by 2008

- **MPRT Forecast**

- Launch module in 2026; Reimbursement in 2027;
- **1.3% adoption by 2030**; 21% by 2035

Financials - Revenue Share

Per-device recurring revenue driven by zero-capex hospital adoption

	2027	2028	2029	2030	2031	2032	Steady-State
RT patients/device/yr	250	250	250	250	250	250	250
MPRT adoption rate	30%	35%	40%	45%	50%	55%	70%
MPRT patients/device/yr	75	88	100	113	125	138	175
MPRT revenue/patient (\$K)	5	5	5	5	5	5	5
MPRT revenue/device/yr (\$K)	375	438	500	563	625	688	875
Revenue share	50%	50%	50%	50%	50%	50%	50%
Revenue/device (\$K)	188	219	250	281	313	344	438
Units sold	5	17	63	151	362	870	1,000
Cumulative units sold	5	22	85	236	598	1,468	10,000
Global RT devices	17,000	17,459	17,930	18,415	18,912	19,422	19,947
MPRT global penetration	0.0%	0.1%	0.5%	1.3%	3.2%	7.6%	50.1%
Total revenue (\$K)	938	4,813	21,250	66,375	186,875	504,625	4,375,000
EBIT	(1,631)	(2,944)	(1,355)	3,605	29,622	102,035	1,093,750

Key Risks & De-risking Strategy

Multi-center clinical validation as the key de-risking milestone

Clinical Adoption

- **Risk:** Slow adoption of new technology
- **De-risking:** multi-center, multi-country trials (Korea, Japan, US)

Regulatory

- **Risk:** Potential classification as a new RT modality
- **De-risking:** Accessory pathway + MR-Linac safety precedent + toxicity-based trials

OEM

- **Risk:** Linac OEM compatibility and warranty concerns
- **De-risking:** OEM-independent accessory deployment, supported by <1% non-interference validation

Funding Request for A2G

Focused on Multicenter Clinical Trials to Accelerate Market Adoption

Funding Plan

	Pre-money value	Raise	Strategic Role	Status
A2G Fund	\$130M (subject to discussion)	\$4M	Institutional lead	Proposal
Physician Consortium	\$130M (aligned)	\$3M	Clinical adoption & KOL alignment	Indicative
National R&D Grant	Non-dilutive	\$20M	De-risk development & regulation	Submitted

Use of Funds

Category	Expected Outcomes	Allocation
Clinical Trials	Multicenter trials for skin/mucosal toxicity reduction	45%
Regulatory	Regulatory-grade safety and clinical evidence	20%
Manufacturing	Pilot manufacturing scale-up	15%
Business Development	Global partnerships and market development	10%
Working Capital	HR, operations, contingencies	10%

Comparable-Based Exit Potential

Exit Upside Benchmarking Based on Comparable Companies

Comparable Companies	Indication	Technology	Pre-approval Funding	Total Funding	Recent Stage	Recent Value
HistoSonics	Liver cancer	Focused ultrasound	\$94M	\$566M	FDA approved, 50 hospitals adoption	M&A, \$2.3B
PROCEPT BioRobotics	Prostate cancer	Water-jet ablation	\$52M	\$349M	FDA approved, \$7M revenue	IPO, \$2B
Auris Health	Lung cancer	Endoscopic surgery	\$512M	\$733M	FDA approved, first-in-human	M&A, \$3.4B+1B
RADEXEL	Prostate, breast, H&N	Magnetic-control RT	\$6M invested + \$7M equity		Pre-approval, Pilot clinical trial	\$200M (KRW 300B)

- Expected Exit Timeline:**

- Current:** Korea regulatory review, pilot trial, pilot production
- 2026-2027:** Multicenter trial, US approval, production scale-up
- 2028-2029:** Global trial, EU/Japan/China approval
- 2031:** M&A or IPO, **\$3-5B value** (Illustrative, based on comparable companies)

1. DCF 평가 결과 Summary				
단위: 백만원		할인율		
		18.17%	19.17%	20.17%
영구성장률	1.00%	321,293	288,346	259,493
	2.00%	338,197	302,538	271,483
	3.00%	357,329	318,486	284,870
2. 상대가치 평가결과 summary				
단위: 백만원		Min	Average	Max
EV/EBITDA		310,319	378,971	497,123
EV/Sales		30,808	384,624	1,048,562

(*) Valuation excludes MPRT-RB and IS

Target Problem

Solution: MPRT-SD

Team & Execution

Market & Business

Expansion: MPRT-IS & MPRT-RB

Product Pipeline

Magnetic Precision Radiation Therapy, MPRT



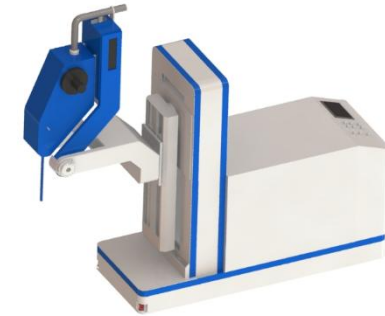
MPRT-SD

Modular MPRT for Standard RT
Fast Adoption & Near-term Revenue



MPRT-IS

Integrated MPRT System
Long-term Differentiation



MPRT-RB

Mobile MPRT Robotic System
Next-Generation Platform

MPRT-IS, Plan

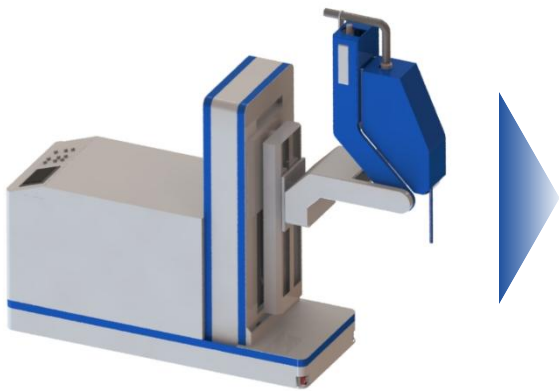
Integrated MPRT System: MPRT-SD + RT machine (National R&D Grant Target)

사업명 사업목표	범부처 첨단 의료기기 연구개발사업 첨단 의료기기 개발을 통한 글로벌 신시장 선점 및 보건안보 대응역량 확보	범부처 첨단 의료기기 연구개발사업 _ B. 사업 주요내용 1 내역 글로벌 플래그십(Flagship) 의료기기 개발
목적 특징 내용	최첨단 기술을 통해 기존 의료기기 성능의 한계를 극복하여 글로벌 의료기기 시장 선점 및 의료현장 내 미충족 수요를 해결하는 '게임체인저 의료기기' 개발 도전적·혁신적 R&D를 통해 의료기기 한계극복 미션* 수행 지원 *도전적 난제 해결 미션 수행: 세계 최초 의료기기 개발, 세계 최고 난이도 의료기기 개발 국가 보건의료 정책목표 달성을 위한 미션에서 주요 난제를 도출하였으며 해결을 위한 플래그십 의료기기 개발 6대 프로젝트 추진	Mission 세계최고 난이도 의료기기
프로젝트명 정상조직 피해 최소화 방사선 암치료기	프로젝트 주요내용 ● 지원기간 : 2026년 ~ 2032년(7년) ● 지원금액 : 290.25 억 원 ● 주요내용 : 방사선으로 인한 면역 체계 저하를 줄이는 초고선량을 방사선 암치료기와 고정밀 중앙추적 및 치료계획 SW 개발 	



MPRT-RB, Plan

Compact Robotic RT System: Pencil beam scanning FLASH-RT



MPRT-RB



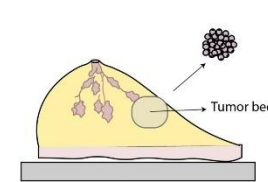
Melanoma



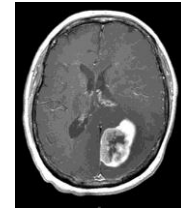
Lymphoma



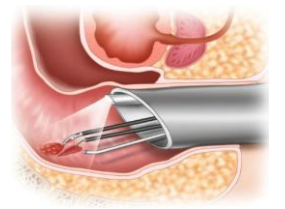
Keloid



Breast



Brain



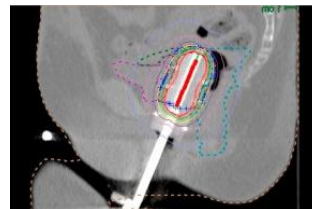
Rectum



Larynx



Oral cavity



Uterus



Pancreas, liver, sarcoma



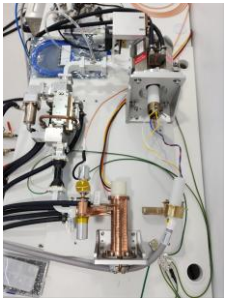
Minimally invasive surgery

Primary indications:
Skin & mucosal tumors

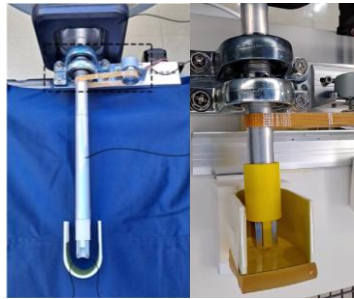
Future indications:
Intraoperative RT

MPRT-RB, Plan

Component Development Completed; Prototype in Progress



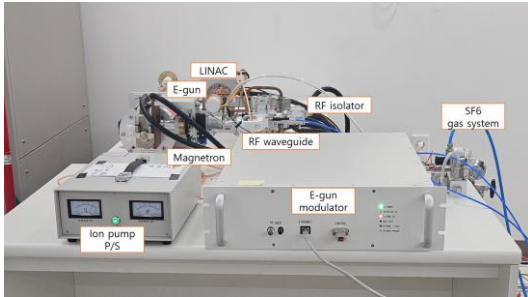
Linear accelerator



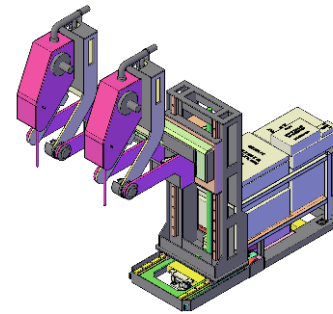
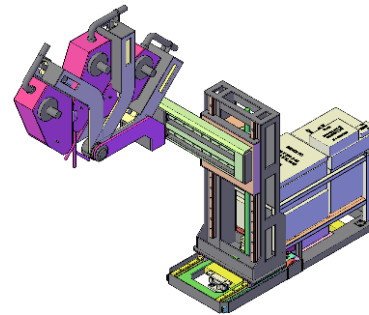
Electron beam output test



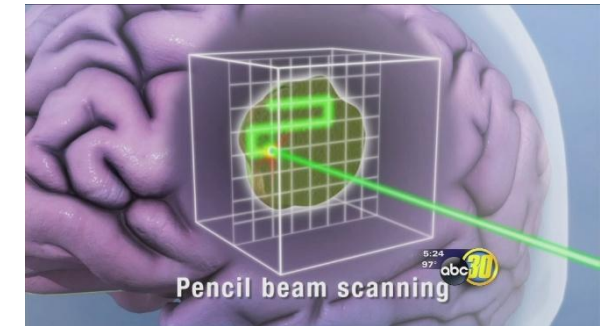
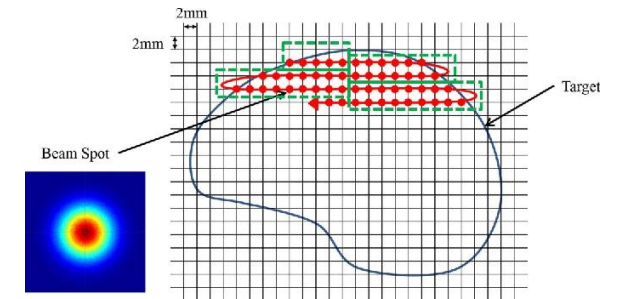
Robotic drive unit



RF, vacuum, and power supply units

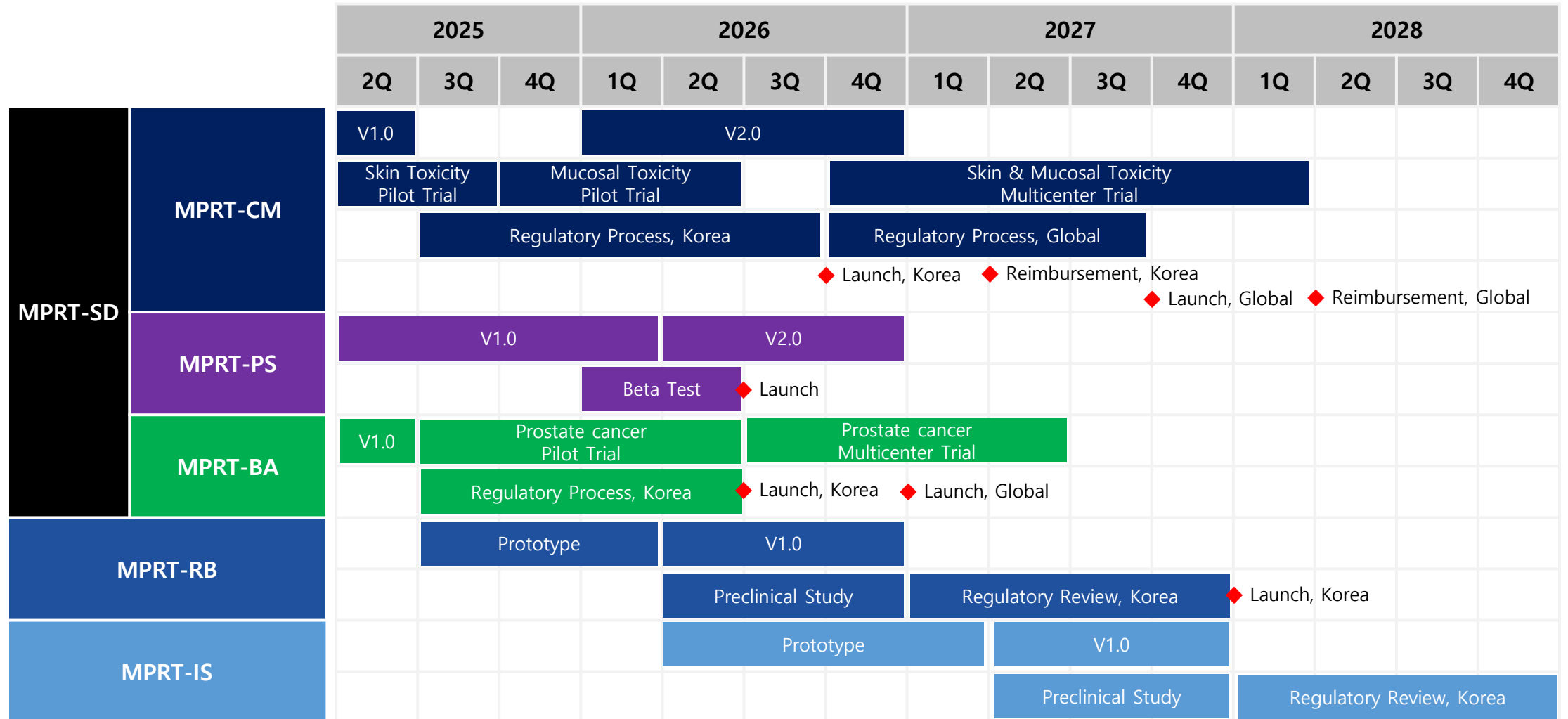


Prototype design



Beam scanning technique

MPRT Platform Roadmap



Transforming Radiation Therapy with Magnetic Precision

Contact: info@radexel.com

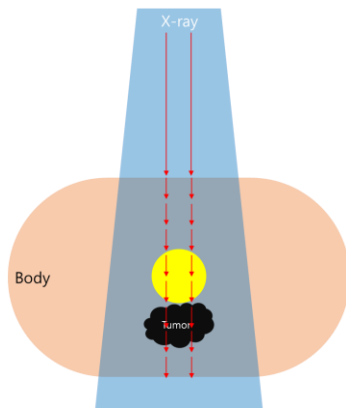
Appendix

MPRT-SD, R&D

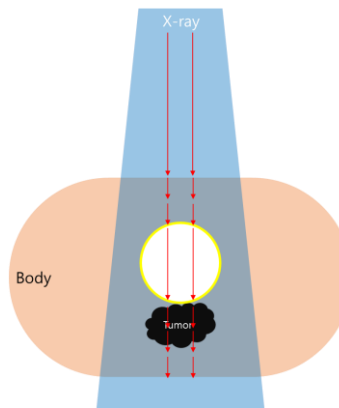
MPRT-BA (Balloon applicator): Creates Space for Magnetic Field Control

Inserted similarly to existing rectal balloon; no new workflow required

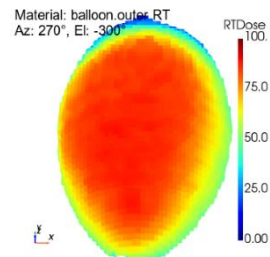
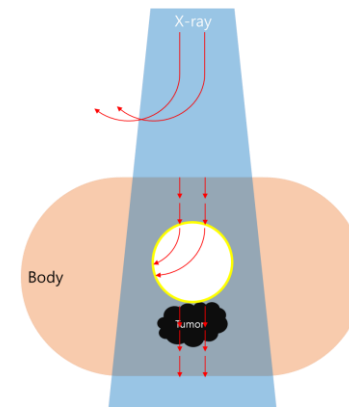
RT



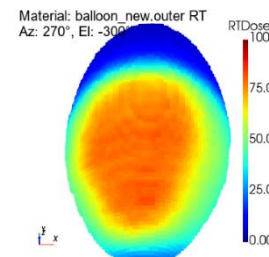
RT + Applicator



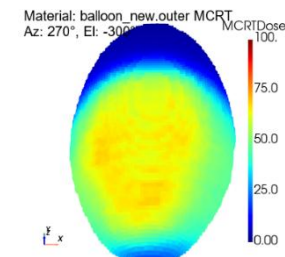
MPRT + Applicator



10% Dose Reduction

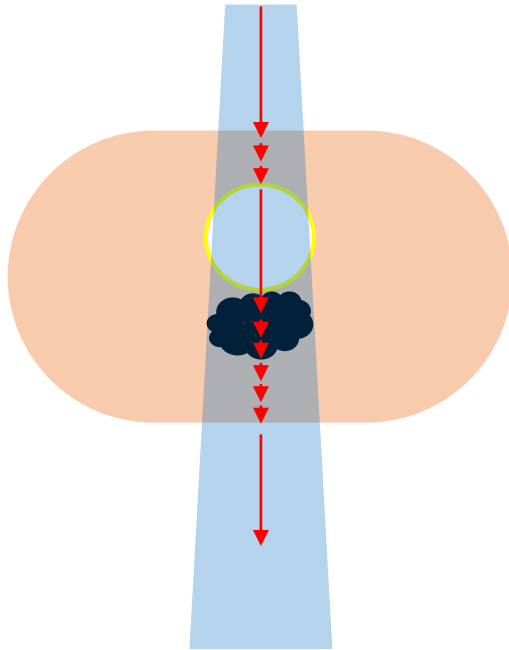


20% Dose Reduction

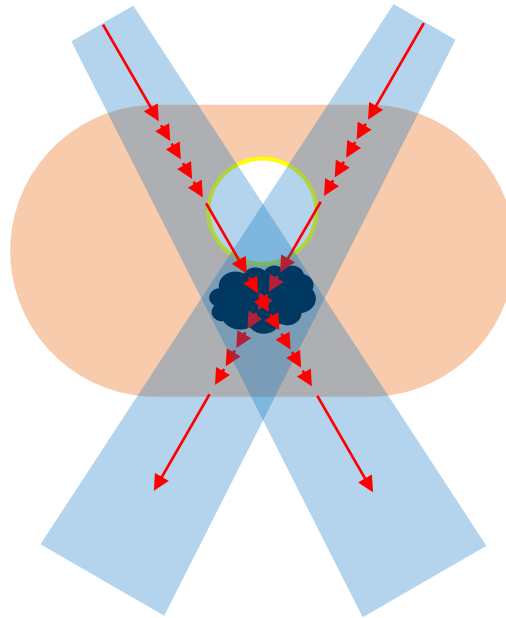


Confidential

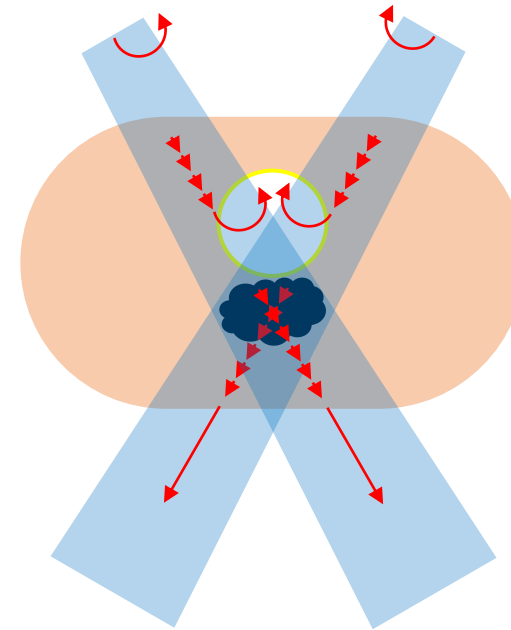
MPRT-SD Mechanism



2D-RT



IMRT



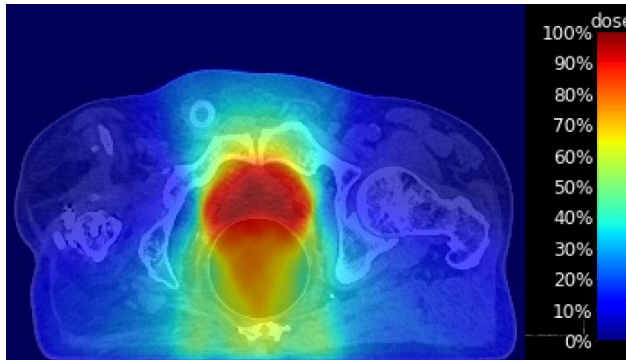
MPRT

MPRT-SD, R&D

MPRT-PS (Planning SW): Enables Treatment Planning with Magnetic Field Integration

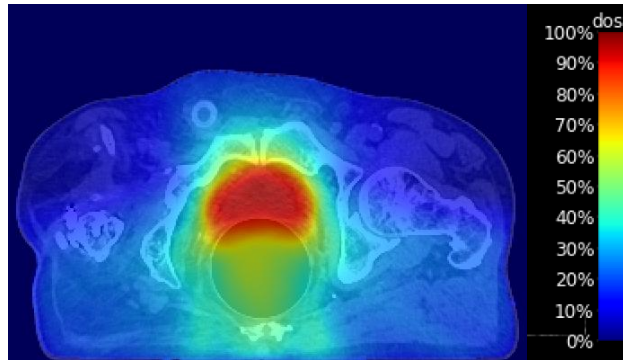
(*) Radiation therapy requires dose-based simulation planning

Conventional Simulation



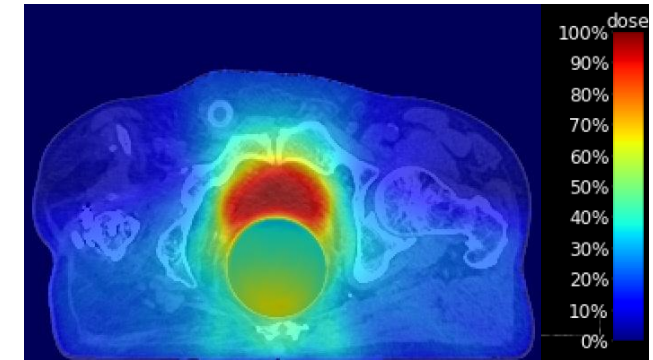
- Inaccurate in heterogeneous tissues (e.g., skin, body cavity)
- Fast calculation (~1 min)
- Not applicable for MPRT

Monte Carlo Simulation



- Gold standard for dose calculation
- Too slow for clinical use (~8 hrs)
- Supports MPRT dose calculation

Fast Monte Carlo for MPRT



- **AI-driven dose prediction**
- **50x faster Monte Carlo simulation (8 hrs → 10 mins)**
- **Supports MPRT dose calculation**

Preclinical Validation

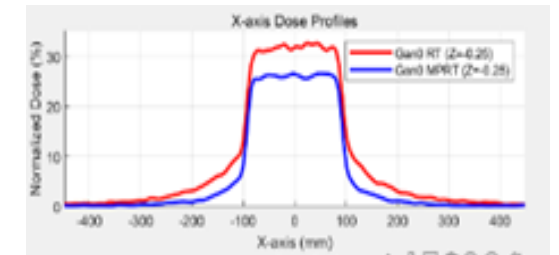
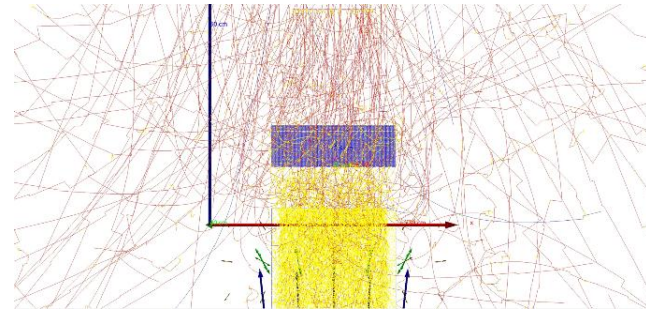
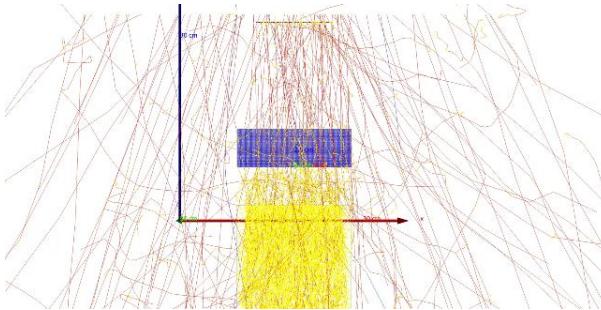
Particle Simulation: ~20% dose reduction at the tissue interface

RT

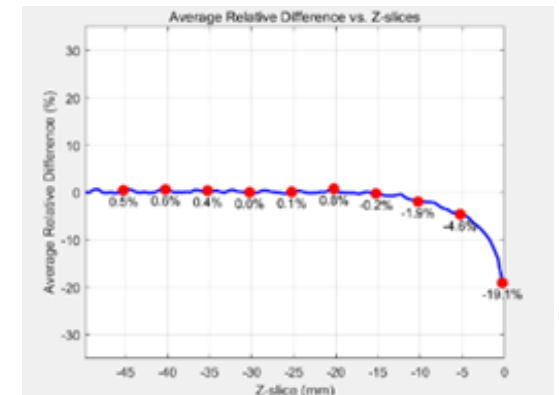
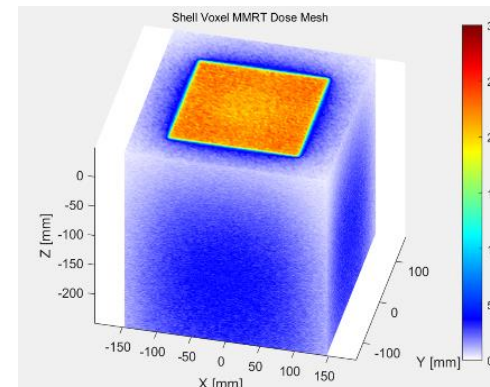
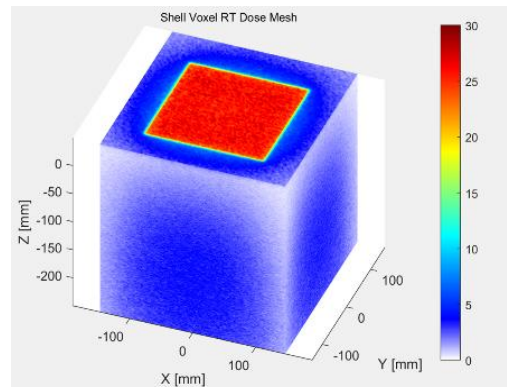
MPRT

Difference

Electron
Diffraction



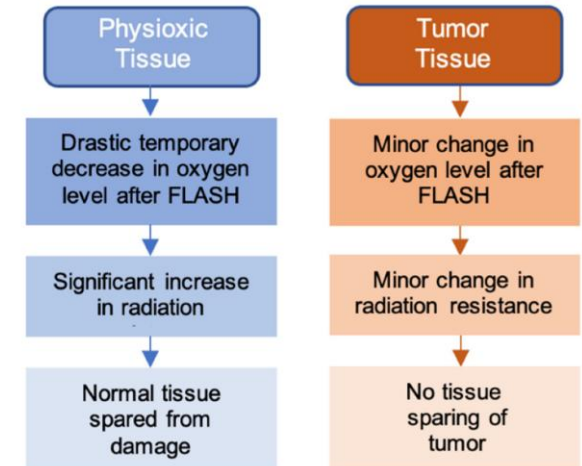
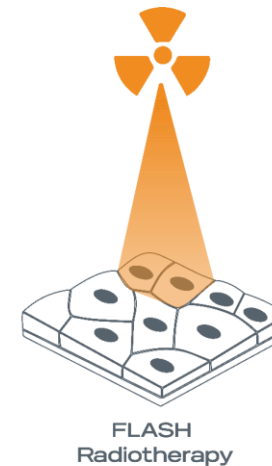
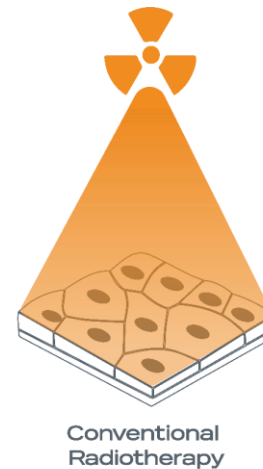
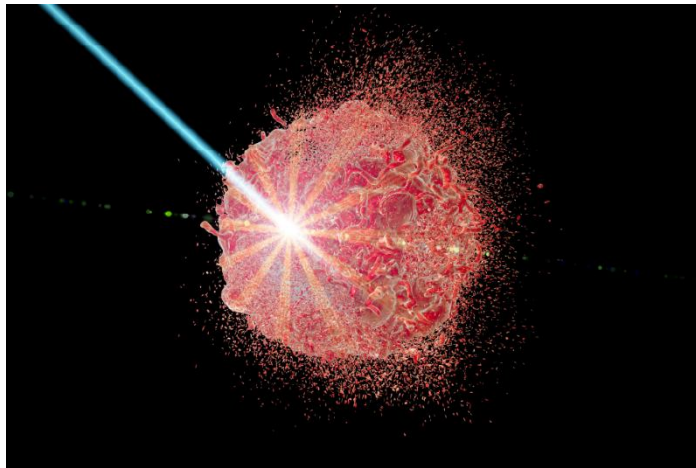
Radiation
Dose



Confidential

MPRT-RB, Target Problem

FLASH-RT: Next generation RT with significant normal-tissue sparing

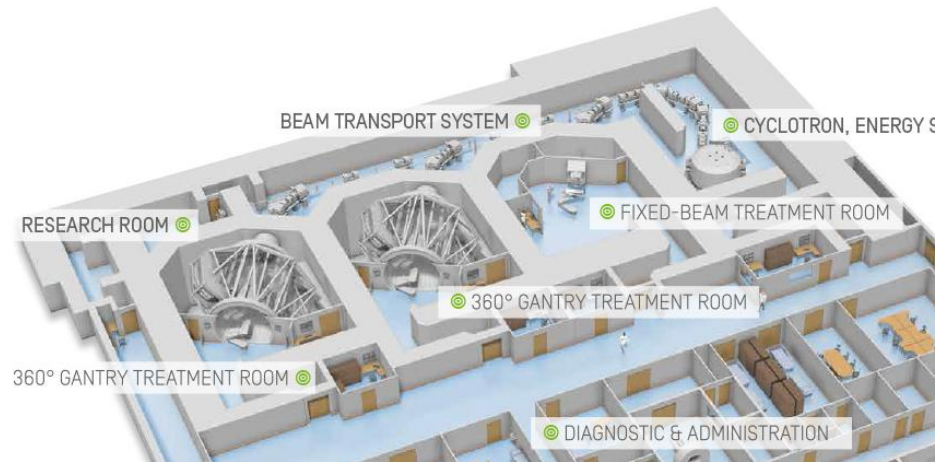


FLASH-RT delivers ultra-high dose-rate radiation using narrow pencil-beam scanning, achieving comparable tumor control while reducing damage to surrounding normal tissue.

MPRT-RB: Technical Concept

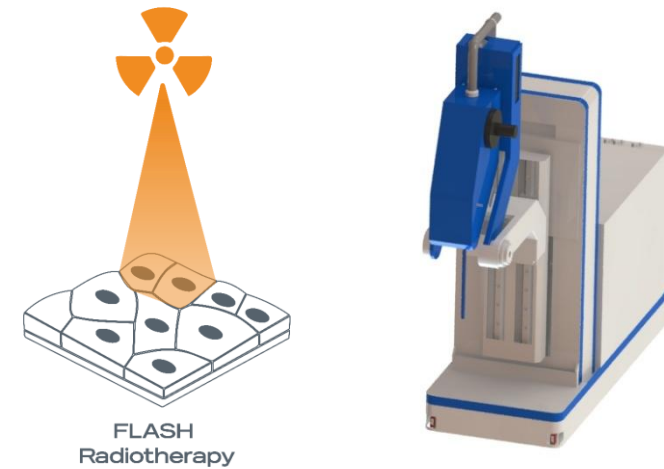
Practical Alternative to Proton-FLASH

Proton-FLASH (under-development)



- Proton therapy has been the primary platform to develop the FLASH effect.
- Requires **large, complex, and expensive facilities (\$30-100M)**
- Difficult to achieve FLASH with precise and stable spot delivery
- → Difficult for most hospitals to adopt

MPRT-RB



- Suppresses electron beam scattering → enables FLASH dose delivery
- Magnetic beam trajectory control → accurate dose delivery
- **Compact and cost-efficient (\$1.5M)** → practical for standard clinics

Q&A

- **FDA 인허가 문제 없나?**

- Add-on Accessory 전략
 - OEM-independent accessor임을 강조 (예, 6D couch, SGRT, immobilization device, IGRT 보조 장비, collimator 등)
- MR-Linac의 Regulatory Precedent 활용
 - RT와 자기장 공존이 치료에 영향 없음
 - MPRT는 MR-Linac 대비 1/10 수준 세기의 자기장 사용
- Risk Analysis Report
 - IEC 60601-2-1, 60601-2-68 등 RT safety norm 기반
 - Magnetic field가 gantry·MLC·EPID·dose monitor에 미치는 영향을 정량화
- Mechanism-driven broad RT accessory claim
 - Skin/mucosal dose reduction accessory for RT
 - Not by separate disease indications
- Clinical KOL 네트워크 활용
 - 7개병원 Multi-center Real-world Data Registry → ASTRO/ESTRO 발표 → KOL endorsement 문서 확보

Q&A

- **Linac OEM (Varian/Elekta)과 충돌 문제 없나?**
 - 기술적
 - 기존 장비와 전기적, 소프트웨어적 연결 없음. → 기존 장비에 직접 영향, 의존 없음
 - 표준적인 QA과정을 통해 종양 치료 선량 보존 확인 완료
 - Tech Report 제공 계획: Radixel Compatibility Validation with Linac Model X/Y/Z
 - 탈착식 제품 → 모듈 제거 시 기존 장비 성능 그대로 사용 가능
 - 식약처 인허가
 - 별도 의료기기로 인허가 진행 중
 - 기존 장비에 대한 영향, 치료 선량 영향 검증 요구 → QA 보고서 통해 대응 중
 - Varian 본사
 - 3rd party에 대한 공식 인증은 없음 (유저가 판단해서 사용하는 것)
 - Collimator, bolus, compensator 등 beam characteristics에 영향 있는 3rd party 제품 사용 중
 - 3rd party 제품 예시: 6D couch, SGRT, immobilization device
 - OEM과 협력 추진
 - OEM의 희망: 치료 혁신, 추가 매출, 소모품 recurring
 - 초기 임상/인허가 이후 OEM 투자 유치, Co-marketing 또는 Bundle 판매 계약

Key Risks & De-risking Strategy

Regulatory Risk (FDA/CE Approval)

- **Risk**
 - The add-on magnetic field device may be classified by the FDA as a “new RT modality.”
 - Concerns around increased clinical evidence requirements and approval timelines.
- **De-risking Strategy**
 - **Secure Accessory Classification:**
 - FDA Pre-Submission meeting planned for early 2026 → target confirmation of accessory pathway.
 - **Leverage MR-Linac Precedent:**
 - Our magnetic field level is 1/10 of MRI; strong safety rationale already established.
 - **Expand Regulatory-grade Safety Data:**
 - Initial phantom QA demonstrates <1% target dose change. Full validation to be expanded with funding.
 - **Apply Adaptive Clinical Designs:**
 - Replace large-scale RCTs with toxicity-endpoint single-arm approval strategies.

Key Risks & De-risking Strategy

OEM Risk (Varian/Elekta Compatibility)

- **Risk**
 - Potential technical and commercial conflicts with linac OEMs.
 - Concerns regarding warranty and TPS interoperability.
- **De-risking Strategy**
 - **Develop a Non-interference Proof Package:**
 - Validate <1% impact on beam quality, dose rate, and gantry stability
 - Reduce potential OEM concerns.
 - **Establish a Compatibility Certification Program:**
 - Adopt a third-party accessory certification approach, enabling installation without requiring formal OEM agreements.
 - **Propose Mid-term OEM Co-marketing Models:**
 - Offer MPRT-SD/MPRT-RB as upgrade platforms to OEMs seeking differentiated products
 - Provide software/consumables revenue-sharing options.

Key Risks & De-risking Strategy

Clinical Adoption Risk

- **Risk**
 - Expanding indications may increase clinical cost and development time.
 - Need to establish trust and credibility among physicians and hospitals for a new technology.
- **De-risking Strategy**
 - **Mechanism-driven Clinical Strategy:**
 - Instead of indication-specific claims (breast/prostate), pursue a broad mechanism-based claim: "Reduction of skin & mucosal toxicity."
 - **Operate a KOL Clinical Consortium (7 leading hospitals):**
 - Launch multi-center registry → Present at ASTRO/ESTRO → drive peer adoption.

Regulatory Approval, Korea

MPR-SD, Korea: Classified as Class II Radiological Treatment Device Approval Expected in Q2 2026



식품의약품안전처

수신 주식회사 라덱셀, 대표:김태순 귀하 (우 24437 강원특별자치도 춘천시 충혼길52번길 10 3층 302호)
(경유)

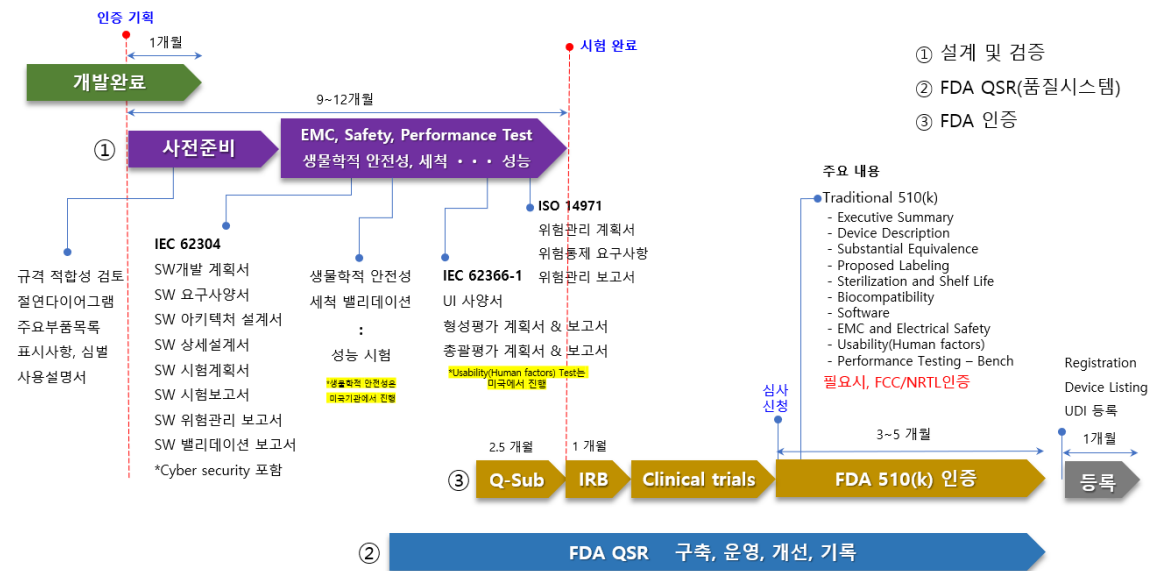
제목 의료기기 해당여부 검토 회신[주식회사 라덱셀(Radexel)]

- 귀하께서 우리처에 신청(접수번호 제20240058206호, 2024. 3. 27.)하신 '의료기기 해당여부 검토'와 관련됩니다.
- 「의료기기법」 제2조제1항에 따라 의료기기란 사람이나 동물에게 단독 또는 조합하여 사용되는 기구·기계·장치·재료·소프트웨어 또는 이와 유사한 제품으로서 아래의 어느 하나에 해당하는 제품을 말합니다(의약품, 의약외품, 의자·보조기 제외).
가. 질병을 진단·치료·경감·처치 또는 예방의 목적으로 사용되는 제품
나. 상해 또는 장애를 진단·치료·경감 또는 보정의 목적으로 사용되는 제품
다. 구조 또는 기능을 검사·대체 또는 변형의 목적으로 사용되는 제품
라. 임신조절의 목적으로 사용되는 제품
- 귀하께서 질의하신 제품이 '방사선치료기의 치료대에 설치되어 내부에 장착된 영구자석에서 발생하는 저강도 자기장으로 방사선 내 전자의 굴절을 유도하여 치료대상의 표면 선량을 낮추는데 사용되는 장치'라면 상기 「의료기기법」 제2조 의료기기 정의에 부합하므로 의료기기에 해당되며, 「의료기기 품목 및 품목별 등급에 관한 규정」(식약처 고시)에 따라 "방사선치료장치"(A13000, 2등급)에 분류될 것으로 사료됩니다.
- 귀하께서 질의하신 내용 및 첨부자료를 바탕으로 해당여부를 판단한 것으로서 질의하신 내용과 사용목적, 작용원리, 성능 등 사실관계가 달라지는 경우에는 달리 판단될 수 있음을 알려드립니다.

- ✓ Uses Magnetic Field ~1/10 of MRI
→ Minimal Impact on Body and Equipment
- ✓ MR-Linac (1.5 T) Already Approved and in Use
- ✓ Minimal impact (<1%) on target dose; primary effect localized at tissue interface.
- ✓ MFDS Indicated **Clinical Trial Exemption**

Regulatory Approval Plan, US

FDA Regulatory Strategy Established



- ✓ Accessory classification pathway (target)
- ✓ Pre-submission meeting in 2026 Q2
- ✓ Safety package: MF phantom QA, beam characterization
- ✓ Clinical plan: toxicity-based single-arm multicenter study

Global Radiation Therapy Equipment

 Status of Radiation Therapy Equipment								
Countries		RT Centres		MV Therapy		Light Ion Therapy		Brachytherapy
155		8,599		17,130		133		3,360
Country	Region	RT Centres	MV Therapy	Light Ion Therapy	kV Therapy	Brachytherapy	Last Update	
Q	Q	Q	Q	Q	Q	Q	Q	
United States of America	Northern America	2237	3892	44	6	638	2024	
China	Eastern Asia	1624	2932	15	124	255	2025	
Japan	Eastern Asia	759	1067	24	0	224	2024	
India	Southern Asia	463	791	1	5	278	2025	
France	Western Europe	168	570	4	2	90	2023	
Russian Federation	Eastern Europe	157	568	4	79	153	2025	
Germany	Western Europe	289	565	7	113	227	2023	
Italy	Southern Europe	204	555	3	12	61	2025	
Brazil	South America	221	370	0	39	128	2023	
United Kingdom	Northern Europe	78	358	7	49	56	2024	
Türkiye	Western Asia	145	300	0	0	31	2025	
Canada	Northern America	52	296	0	14	52	2025	
Spain	Southern Europe	126	262	3	21	135	2024	
Australia	Australia and New Zealand	103	224	0	6	10	2025	
Mexico	Central America	114	207	0	2	66	2025	
Korea, Republic of	Eastern Asia	65	190	2	0	37	2023	
Poland	Eastern Europe	47	173	2	0	24	2023	
Taiwan, China	Eastern Asia	74	156	4	0	19	2024	
Netherlands, Kingdom of the	Western Europe	27	154	3	6	31	2025	
Thailand	South-Eastern Asia	56	154	1	4	25	2025	
Argentina	South America	89	131	0	8	47	2025	
		8599	17134	133	706	3360	Latest:...	

<https://dirac.iaea.org/Query/StatusRTE>
<https://dirac.iaea.org/Query/Countries>

Comparable Market Penetration: IMRT

A Survey of Intensity-Modulated Radiation Therapy Use in the United States

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John C. Roeske, Ph.D.¹
Arno J. Mundt, M.D.^{1,2}

¹ Department of Radiation and Cellular Oncology, University of Chicago, Chicago, Illinois.

² Department of Radiation Oncology, University of Illinois at Chicago, Chicago, Illinois.

BACKGROUND. The objective of this study was to assess the current level of intensity-modulated radiation therapy (IMRT) use in the United States.

METHODS. Three-hundred thirty-three randomly selected radiation oncologists were sent a 13-question survey regarding IMRT use. IMRT users were asked about the number of patients and sites treated, their reasons for adopting IMRT, and future plans for its use. Physicians who did not use IMRT were asked about their reasons for not using IMRT; whether they intended to adopt it in the future; and, if so, their reasons.

RESULTS. One-hundred sixty-eight responses (50.5%) were received. Fifty-four respondents (32.1%) stated that they currently used IMRT. Most IMRT users (79.6%) had adopted IMRT since 2000. Academic physicians were more likely to use IMRT ($P = 0.003$) compared with private practitioners. The percent of physicians using IMRT in practices comprised of 1 physician, 2–4 physicians, or > 4 physicians were 15.4%, 28.4%, and 44.2%, respectively ($P = 0.02$). The most common sites treated were head and neck malignancies and genitourinary tumors. Of the 114 IMRT nonusers, 96.5% planned to use IMRT in the future, with 91.8% planning to use IMRT within 3 years. Among IMRT nonusers, the most common reason cited for not using IMRT was lack of necessary equipment. The most common reasons for adopting IMRT (users) or wanting to adopt IMRT (nonusers) were to improve delivery of conventional doses and to escalate dose.

CONCLUSIONS. Approximately one-third of radiation oncologists in the United States use IMRT. However, this number appears to be growing rapidly. Efforts to ensure the safe and appropriate application of this new technology are warranted.

Cancer 2003;98:204–11. © 2003 American Cancer Society.

KEYWORDS: dose escalation, intensity-modulated radiation therapy, radiation oncologists, survey.

| [Articles](#) | October 2, 2008

Oncology NEWS International

Oncology NEWS International Vol 17 No 10 | Volume 17 | Issue 10

Report finds high rate of adoption of IMRT

DES PLAINES, Illinois—Fully 87% of U.S. radiation oncology sites have adopted intensity-modulated radiation therapy (IMRT), according to the “2007 Radiation Oncology Market Summary Report.”

ABSTRACT: The large majority of U.S. radiation oncology sites are now using this therapy.

DES PLAINES, Illinois—Fully 87% of U.S. radiation oncology sites have adopted intensity-modulated radiation therapy (IMRT), according to the “2007 Radiation Oncology Market Summary Report.”

The report, published by the IMV Medical Information Division, is based on profiles of hospital and free-standing radiation oncology sites throughout the country. Findings reflect responses from 652 sites interviewed during an 18-month period (January 2007–June 2008).

Cost of Comparable Modality: SpaceOAR

SpaceOAR Hydrogel Injection

ESTIMATED
NATIONAL AVERAGE ①

\$15,205

MDSAVE
NATIONAL AVERAGE

\$7,569

Save \$7,636

How Much Does a SpaceOAR Hydrogel Injection Cost?

Purchase a SpaceOAR Hydrogel Injection today on MDsave. Costs range from **\$5,924** to **\$9,031**. Those on high deductible health plans or without insurance can save when they buy their procedure upfront through MDsave. [Read more about how MDsave works.](#)

<https://www.mdsave.com/procedures/spaceoar-hydrogel-injection/d281fbcc>

A SpaceOAR Hydrogel injection costs between \$5,924 and \$7,185, making it largely unaffordable for those without insurance, or those with high deductibles. Even for those *with* health insurance, the insurance company may refuse to pay for the medication. The excuse used by the insurer may be that the gel is a preventative medication rather than a treatment. Considering the fact that SpaceOAR has the potential to prevent further medical damage for a prostate cancer patient, you can **appeal** the denial from your insurance company.

<https://www.scottglovsky.com/insurance-bad-faith/health-insurance-claim-denials/spaceoar-hydrogel-insurance-denial/#~977dc58e-b1c8-41fd-a511-75ded6a55cd6>

RT market data



Benchmark Report

Radiation Therapy
2019

Relative Distribution of Courses of Treatment by Cancer Type and Percent of Sites Performing, 2018

Of the 1,026.7K courses of treatment accommodated in 2018, the top three cancer types include breast, prostate and lung cancer, which account for 25%, 20% and 14%, respectively, of all cancer types treated with radiation therapy. Head & neck cancer accounts for 11% of all cancer types treated, while brain metastases, GYN, brain, and skin cancer account for 22% of cancer types treated, each at 7% or less of the total distribution.

Cancer Types Treated with Radiation Therapy, 2018	% of Courses of Treatment Performed (N = 1,026.7 K)	% of Sites Performing Treatments (N = 2,285)
Breast	25%	93%
Prostate	20%	96%
Lung	14%	94%
Head & Neck	11%	92%
Brain Metastases	7%	91%
GYN	7%	82%
Brain	4%	83%
Skin	4%	83%
All Other	8%	NA
	100%	

Productivity per External Beam Treatment Unit, 2018, as of 2019 Census Survey

Average per Unit	All Sites	Non-Hospitals	Hospitals	<200 Beds	200-399 Beds	400+ Beds
Average Total Patients per EB Unit	250	290	220	215	235	215
Average New Patients per EB Unit	215	245	190	190	205	180
Radiation Therapists per EB Unit	2.9	3.0	2.9	2.7	2.7	3.1

RT market data



[Home](#) / [News](#) / IAEA Directory Provides Data-driven Insights Informing Cancer Care

IAEA Directory Provides Data-driven Insights Informing Cancer Care

For the World Intellectual Property Organization's (WIPO) [Global Innovation Index \(GII\)](#) 2024 which was released in September, DIRAC helped show a [2.7 per cent increase in global radiotherapy equipment](#) from 2022 to 2023. "WIPO's GII Tracker is not only about tracking global innovation investments – that is easy; metrics such as research and development or venture capital are readily available. Documenting global expenditures on this front is our bread and butter," noted Sacha Wunsch-Vincent, Head of Section in WIPO's Department for Economics and Data Analytics and co-editor of GII. "A more interesting and critical challenge for us is to improve our measurement of technology and innovation adoption – i.e. how do available innovations serve to buttress productivity or the everyday lives of citizens, and this in particular in fields such as health and medical technologies."

<https://www.iaea.org/newscenter/news/iaea-directory-provides-data-driven-insights-informing-cancer-care>

Mean RT cost

Comparative Study > J Clin Oncol. 2018 Jun 20;36(18):1823-1830. doi: 10.1200/JCO.2017.75.5371.

Epub 2018 Mar 21.

Comparative Toxicities and Cost of Intensity-Modulated Radiotherapy, Proton Radiation, and Stereotactic Body Radiotherapy Among Younger Men With Prostate Cancer

Hubert Y Pan¹, Jing Jiang¹, Karen E Hoffman¹, Chad Tang¹, Seungtaek L Choi¹, Quynh-Nhu Nguyen¹, Steven J Frank¹, Mitchell S Anscher¹, Ya-Chen Tina Shih¹, Benjamin D Smith¹

Affiliations + expand

PMID: 29561693 PMCID: PMC6008105 DOI: 10.1200/JCO.2017.75.5371

Abstract

Purpose To compare the toxicities and cost of proton radiation and stereotactic body radiotherapy (SBRT) with intensity-modulated radiotherapy (IMRT) for prostate cancer among men younger than 65 years of age with private insurance. **Methods** Using the MarketScan Commercial Claims and Encounters database, we identified men who received radiation for prostate cancer between 2008 and 2015. Patients undergoing proton therapy and SBRT were propensity score-matched to IMRT patients on the basis of clinical and sociodemographic factors. Proportional hazards models compared the cumulative incidence of urinary, bowel, and erectile dysfunction toxicities by treatment. Cost from a payer's perspective was calculated from claims and adjusted to 2015 dollars. **Results** A total of 693 proton therapy patients were matched to 3,465 IMRT patients. Proton therapy patients had a lower risk of composite urinary toxicity (33% v 42% at 2 years; $P < .001$) and erectile dysfunction (21% v 28% at 2 years; $P < .001$), but a higher risk of bowel toxicity (20% v 15% at 2 years; $P = .02$). Mean radiation cost was \$115,501 for proton therapy patients and \$59,012 for IMRT patients ($P < .001$). A total of 310 SBRT patients were matched to 3,100 IMRT patients. There were no significant differences in composite urinary, bowel, or erectile dysfunction toxicities between SBRT and IMRT patients ($P > .05$), although a higher risk of urinary fistula was noted with SBRT (1% v 0.1% at 2 years; $P = .009$). Mean radiation cost for SBRT was \$49,504 and \$57,244 for IMRT ($P < .001$). **Conclusion** Among younger men with prostate cancer, proton radiation was associated with significant reductions in urinary toxicity but increased bowel toxicity at nearly twice the cost of IMRT. SBRT and IMRT were associated with similar toxicity profiles; SBRT was modestly less expensive than IMRT.

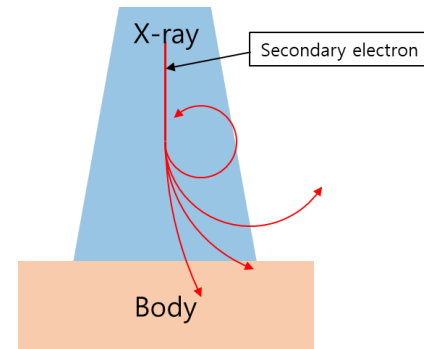
Technical Background

Magnetic Control Can Adjust Electron Paths and RT Dose Distribution

$$E = mc^2 = \gamma m_0 c^2$$

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$r = \frac{mv}{qB}$$



Electron kinetic energy (MeV)	Electron velocity (% /light speed)	Relativistic electron mass (/rest mass)	Electron penetration depth (cm)	Magnetic field strength (T)	Electron trajectory radius (cm)
0.0	0.0	1.0	0.0	0.1	1.7
0.1	54.8	1.2	0.03	0.1	2.0
1.0	94.1	3.0	0.3	0.1	5.0
2.0	97.9	4.9	0.7	0.1	8.3
2.0	97.9	4.9	0.7	0.2	4.1
6.0	99.7	12.7	2.0	0.1	21.2
6.0	99.7	12.7	2.0	0.2	10.6

Team: Experts in Each Field

Management



김태순, MD, CEO, Co-founder
서울대 의학
(전) AI 신약개발사 CEO
경영 총괄



정누리현, MD, Co-founder
방사선종양학
(전) 서울아산병원
R&D, 임상 총괄



정수현, MBA
고려대 경영학
재무, 인사



이도영
경영학
회계, 관리

Technical Strategy



김태훈, PhD
원자력공학
(전) 서울아산병원
기술전략, 입자 시뮬레이션



윤영준, MD
방사선종양학
학술 데이터



RA/QA

이수민, MD
방사선종양학
(전) 서울아산병원
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RA



왕인영
경영학
QA

Robotics



방정배, PhD
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시스템 설계



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장비 설계



김재현
기계공학
로봇장비 설계 8년
장비 설계



김원준
전기공학
전장, 제어

Software



남승희, PhD
컴퓨터공학
(전) 포항공대 가속기연구소
시스템 제어



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컴퓨터공학
(전) 중이온가속기연구소
소프트웨어 개발



이유진
간호학, 풀스택
SW 개발



염민지
컴퓨터공학, 풀스택
SW 개발

MPRT-RB, Target Problem

Intraoperative RT reduces side effects but is not compatible with robotic surgery.



Surgery of the past

Compatible



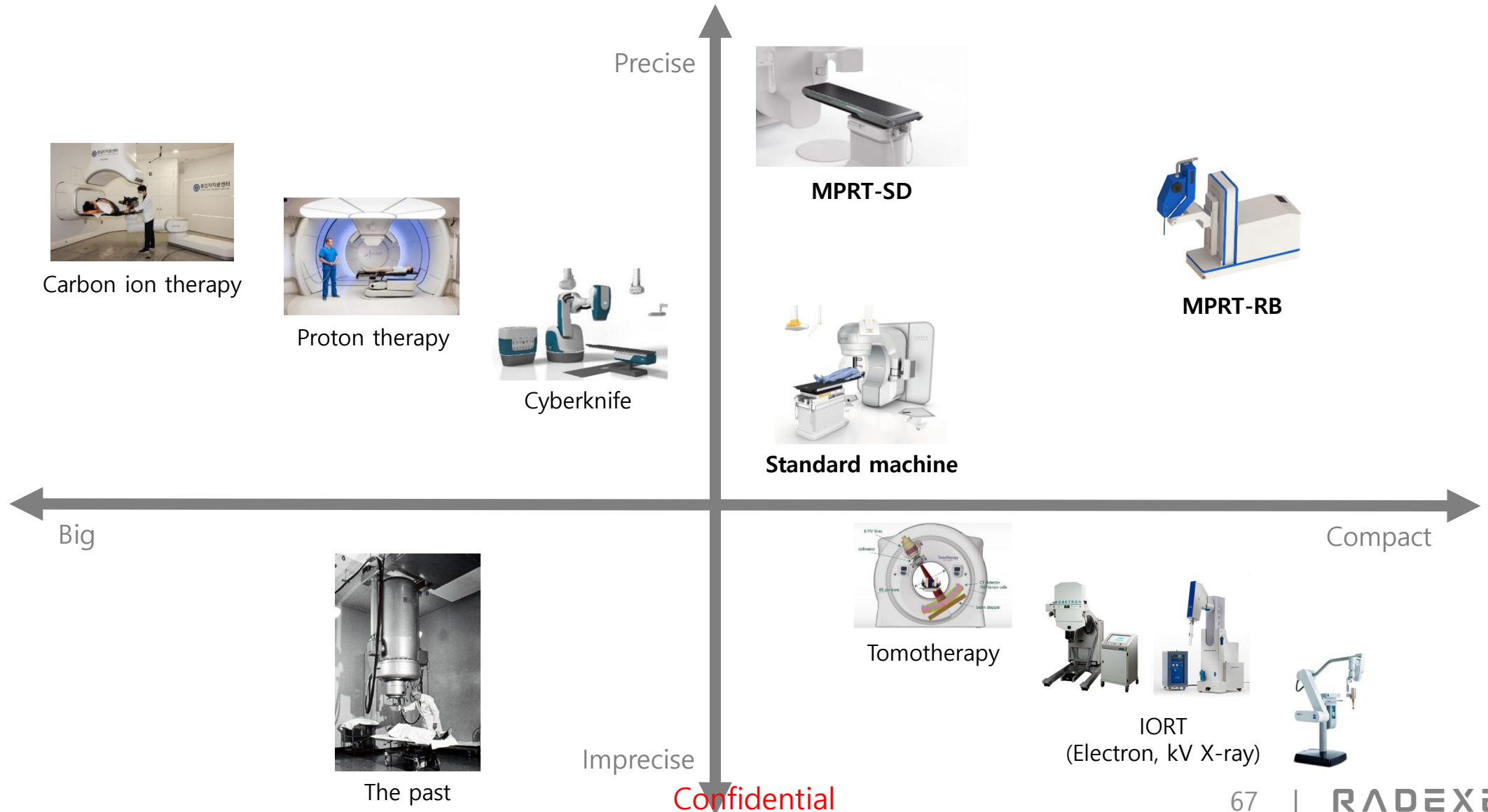
Current RT

NOT compatible



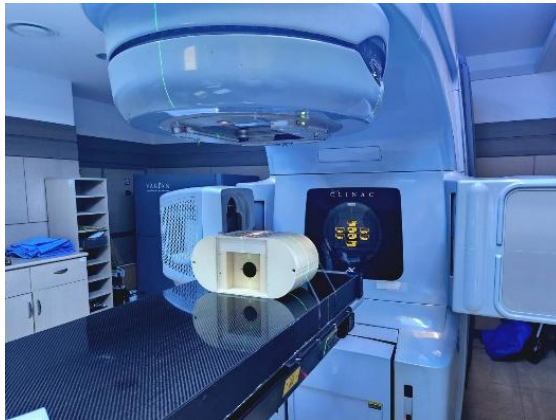
Minimal Invasive Surgery

Competition & Advantage

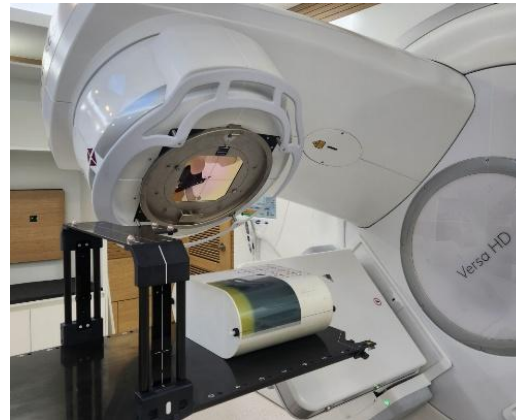


MPRT SD, R&D

MPRT Module: Compatible with Major RT Systems



Siemens/Varian (US)



Elekta (Sweden)



Shinva (China)
For Research Only

Team

Management



Kim Tyson, MD, CEO, Co-founder
Former CEO of AI drug startup
Management



Jung Nuri Hyun, MD, Co-founder
Radiation Oncologist
R&D, Strategy



Jung Su-hyun, MBA
Business Administration
Finance, HR



Lee Do-young
Business Administration
Accounting, Business support

Technical Strategy



Kim Tae-hoon, PhD
Nuclear Engineering
Strategy, Particle simulation



Yoon Young-jun, MD
Radiation Oncologist
Data research, UX



RA/QA

Lee Su-min, MD
Radiation Oncologist
RA/QA, Clinical trial



Song Ye-eun
Medical engineering
RA



Wang In-young
Business Administration
QA

Robotics



Bhang Jung-bae, PhD
Accelerator physics
Particle accelerator design



Nam Seung-hee, PhD
Computer Engineering
Particle accelerator control



Kim Sung-jun
Mechanical engineering
Medical device design



Kim Jae-hyun
Mechanical engineering
Robotics design



Kim Won-jun
Electrical engineering
Electrical control

Software



Lee Sang-gil, PhD
Computer Engineering
Software development



Park Geun-woo
Statistics
Software development



Lee Yu-jin
Nursing
Software development



Yeom Min-ji
Computer Engineering
Software development

Team: Experts in Each Field

Management



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R&D, 임상 총괄



정수현, MBA
고려대 경영학
재무, 인사



이도영
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회계, 관리

Technical Strategy



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송예은
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방사선 의료장비 RA 12년
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장비 설계



김재현
기계공학
로봇장비 설계 7년
장비 설계



김원준
전기공학
전장, 제어

Software



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소프트웨어 개발



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간호학, 풀스택
SW 개발



염민지
컴퓨터공학, 풀스택
SW 개발

국가과제 선정

총 45억원 규모의 국가과제 선정으로 동 분야 최고수준의 역량 입증

2022~2023



자기장-방사선 융합 암 치료를 위한 데이터 기반 선량 조절 기술개발

5억원

2023~2025



데이터 기반 자기장-방사선 융합 암치료 상용화 및 실증

20억원

2025~2028



차세대 방사선 치료 수용성을 향상시키는 혁신형 피부선량 저감모듈 개발 및 실증

20억원

Expansion Strategy

Key Prerequisite: Presenting Clinical Trial Results at Major Conferences Focus on Highlighting Successful Clinical Cases

- 17 hospitals within the Radixel network, equipped with 43 RT devices.

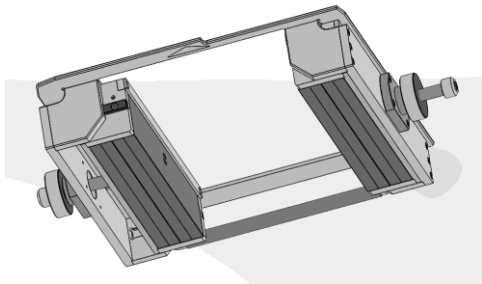
- Target 3 conferences: KOSRO (Korea), U.S. (ASTRO), and Europe (ESTRO).



- Easier data collection on MPRT patients among early customers.
- Radixel will provide full support to facilitate analysis of MPRT patient outcomes.
- Key driver for adoption: credible data presented at major conferences.
- Evidence of improved MPRT patient outcomes and enhanced hospital profitability will accelerate adoption.

MPRT-SD, R&D

MPRT-HM & MPRT-CM: Modular Magnetic Field Generator



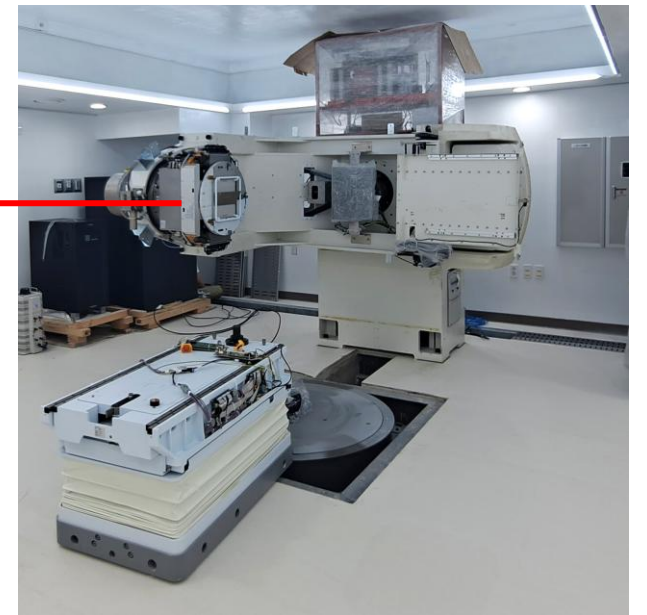
MPRT-HM (Head Module)



MPRT-CM (Couch Module)



MPRT-SD



X-ray RT device

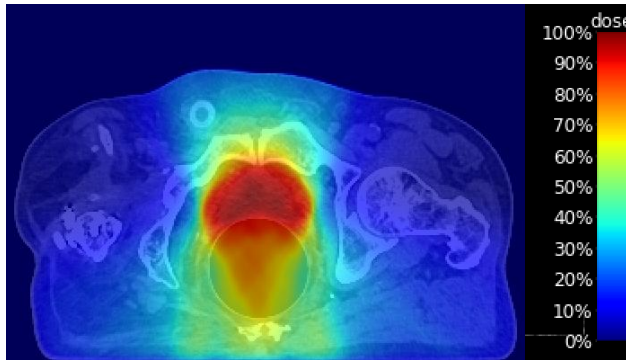
MPRT-SD, R&D

MPRT-PS: Planning SW

Enables Treatment Planning with Magnetic Field Integration

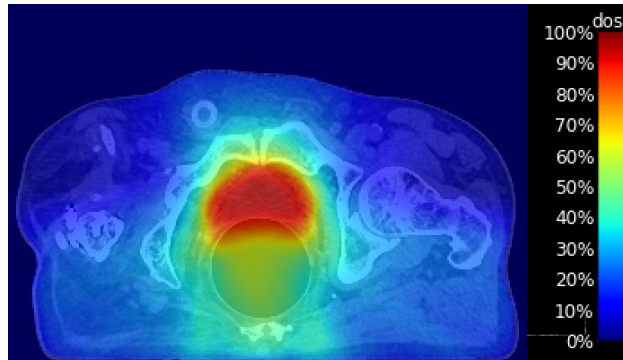
(*) RT requires dose-based simulation planning

RT Conventional



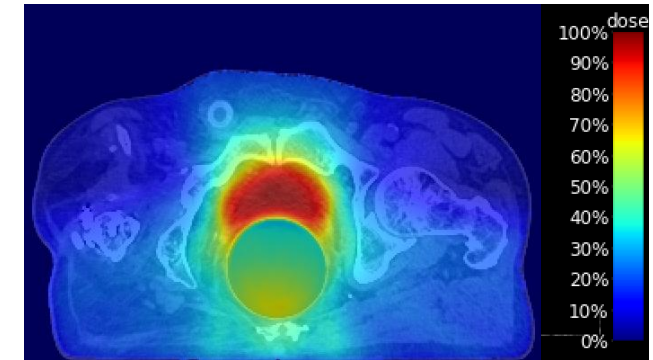
- Based on Analytical Models
- Inaccurate in Heterogeneous Tissues (e.g., Skin, Lung)
- Fast Calculation, but Not Applicable for MPRT

RT Monte Carlo



- Based on Probabilistic Models
- Gold Standard for Dose Calculation
- Supports MPRT Dose Calculation but Too Slow for Clinical Use

MPRT Fast Monte Carlo

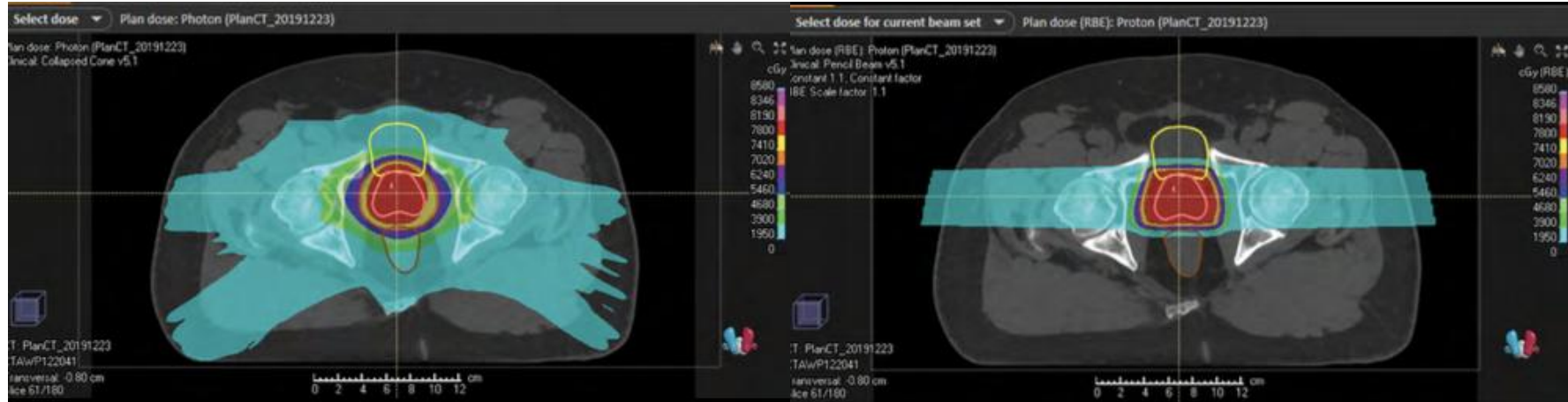


- Enables Accurate MPRT Dose Calculation
- Faster than Traditional Simulation
- Requires 20 Min per Patient

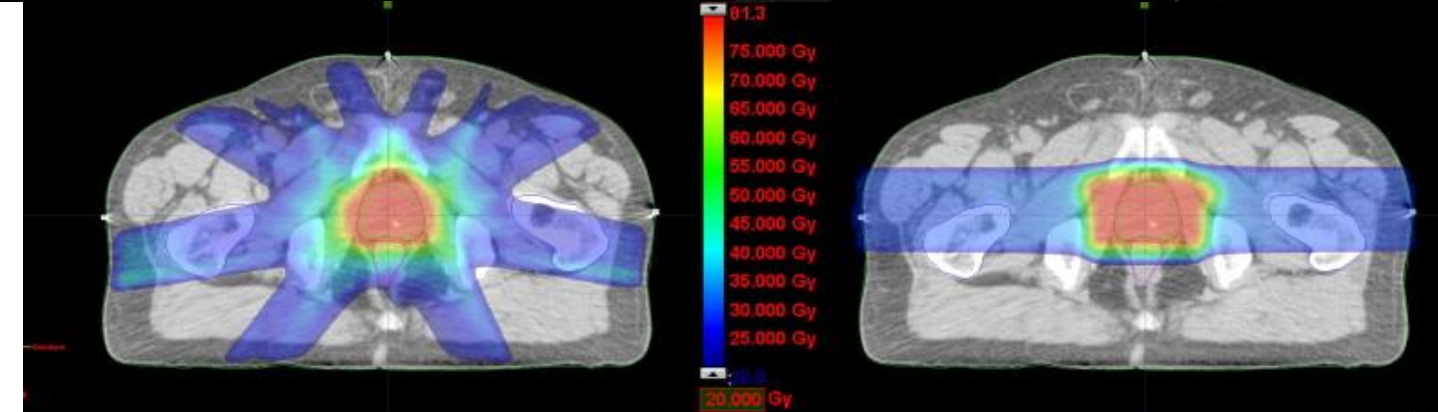
X-ray RT

Particle Therapy

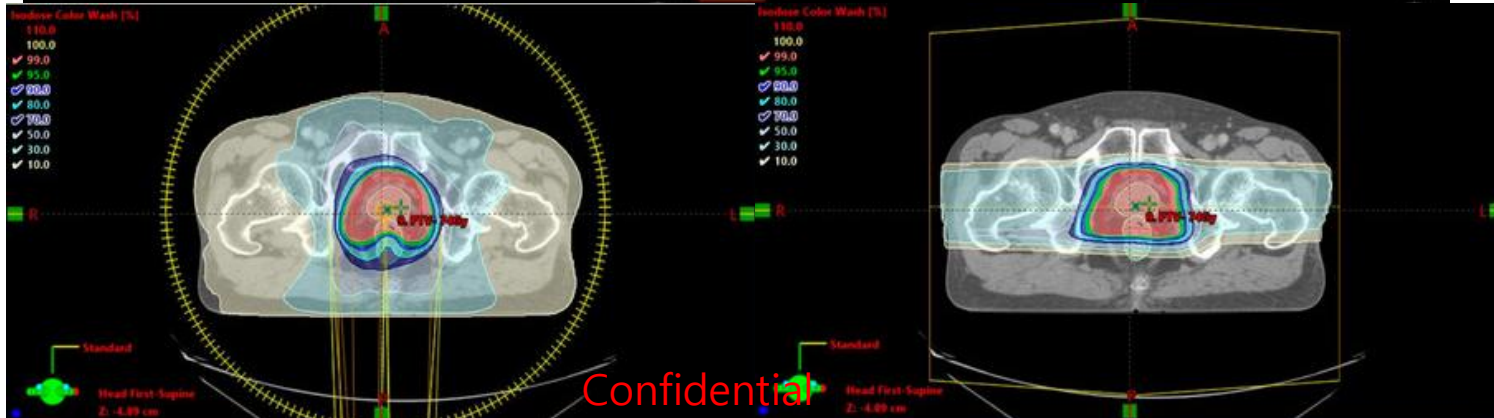
Johns Hopkins
Cancer Center



M. D. Anderson
Cancer Center



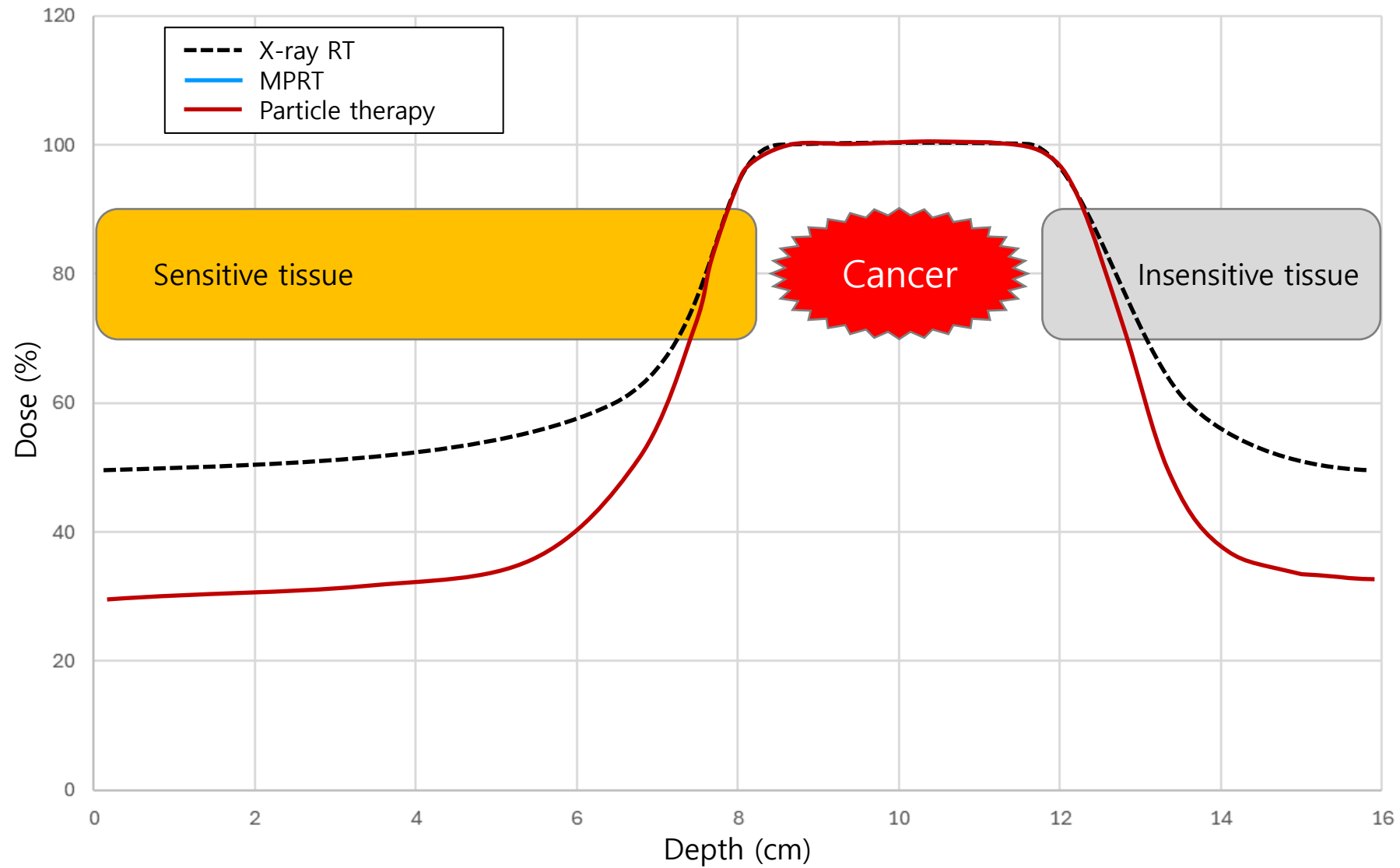
HKSH, Hong Kong



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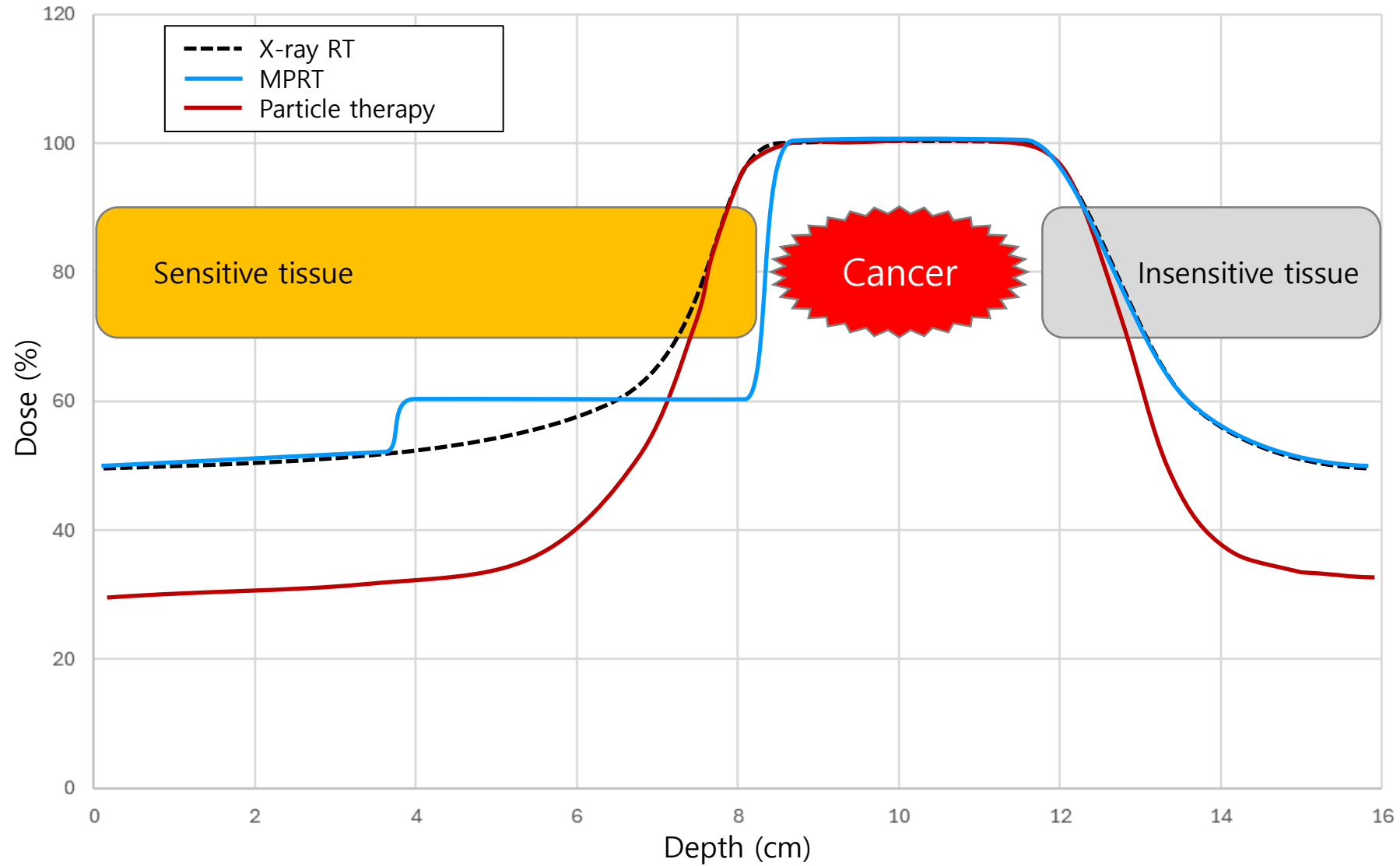
RADEXEL

MPRT SD, mechanism



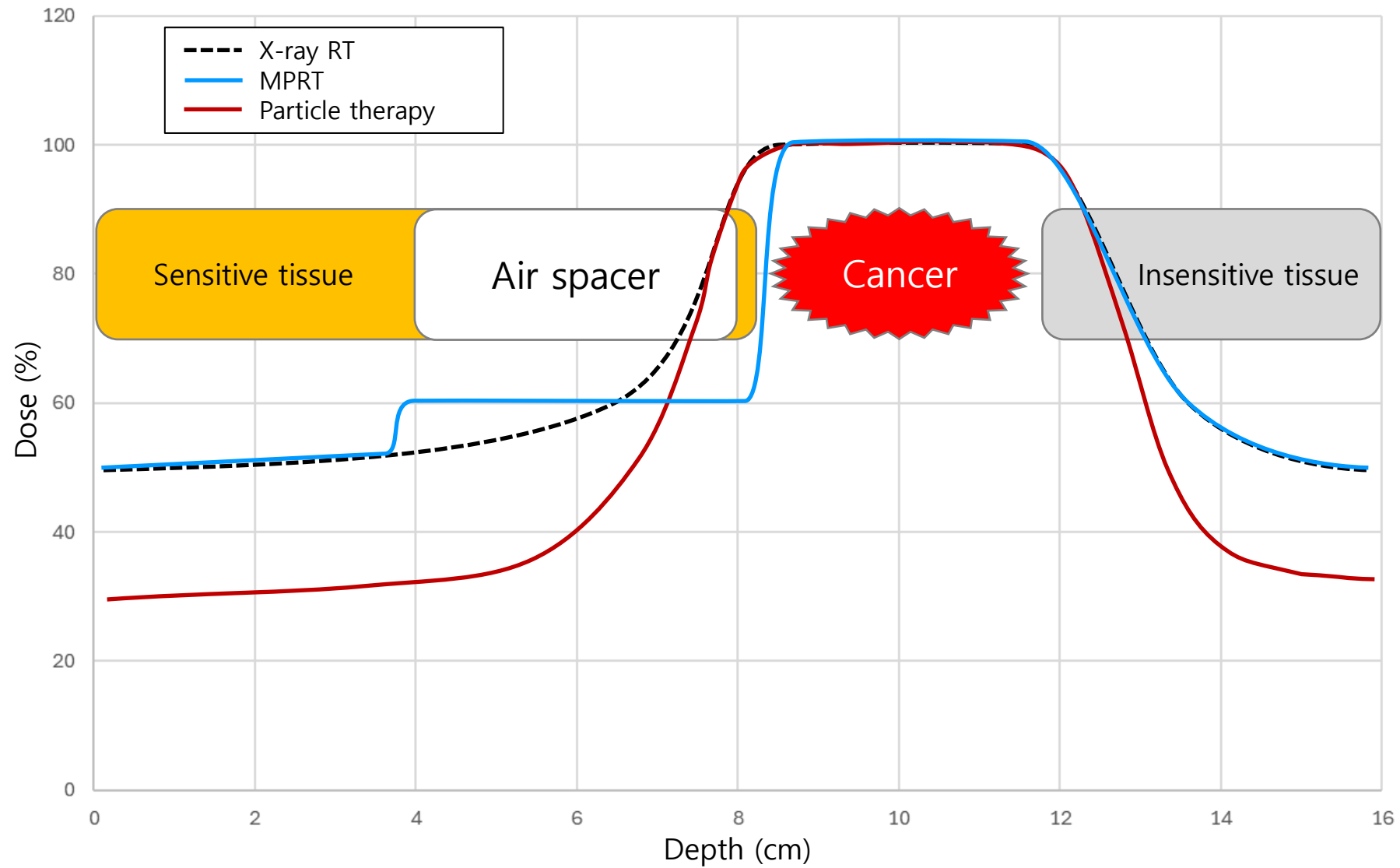
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MPRT SD, mechanism



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MPRT SD, mechanism



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